



R&D Project: System-Level Modeling of a 5 GHz Wireless Transceiver

**In collaboration with [IHP GmbH - Leibniz Institute for High Performance
Microelectronics](#)**

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Abstract

This research and development project is carried out in collaboration between IHP GmbH - Leibniz Institute for High Performance Microelectronics and Aarhus University during the spring semester of 2026. The project corresponds to a 10 ECTS workload and focuses on system-level behavioral modeling of a 5 GHz IQ transceiver for IEEE 802.11a WLAN applications. The objective is to derive realistic design requirements for a phase-locked loop (PLL) by linking system performance to underlying architectural parameters.

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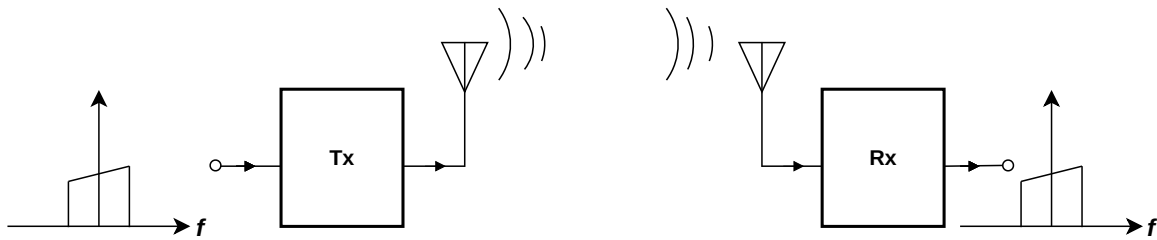
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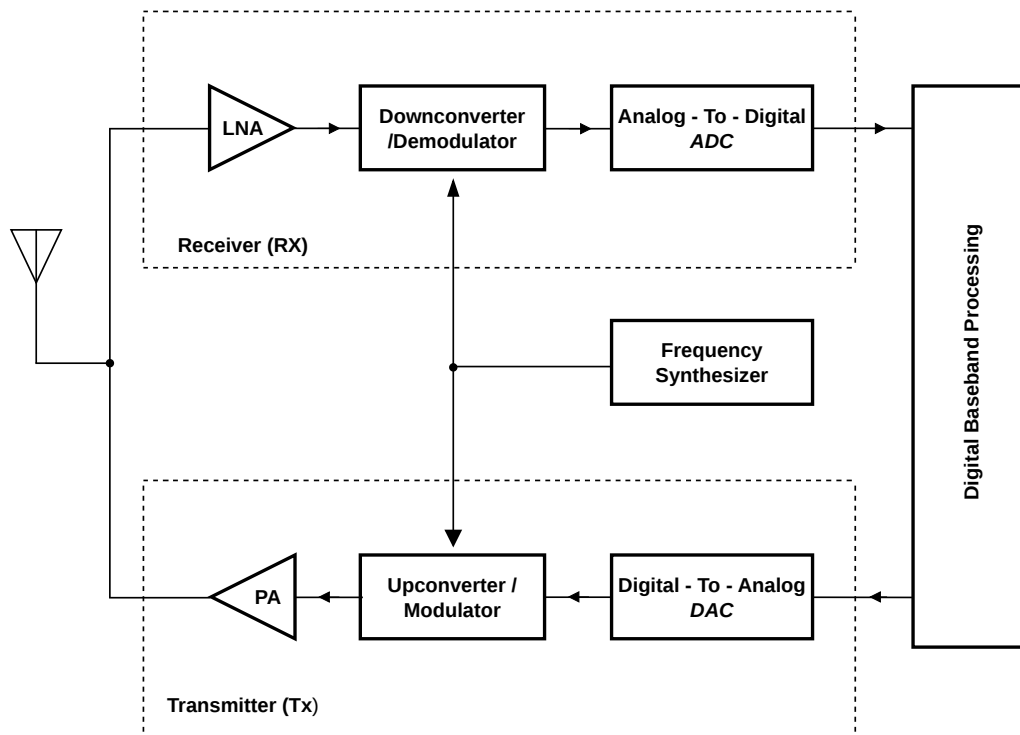
1 | Introduction

Modern wireless communication systems, including WLAN, 5G, and satellite links, are required to meet stringent performance metrics in order to support high data rates and comply with regulatory constraints. These systems are typically implemented using radio-frequency (RF) transceivers composed of discrete functional blocks such as low-noise amplifiers (LNAs), mixers, power amplifiers (PAs), and phase-locked loops (PLLs). In practice, each component requires careful design, and achieving proficiency in RF circuit design can take years. While an individual component within an RF transceiver may exhibit near state-of-the-art performance, the overall system performance can still be poor due to the interaction of multiple factors, including component nonlinearities, impedance mismatches, and external noise sources. Therefore, it is important to analyze such systems at a holistic level and derive system-level specifications that can be translated into component-level requirements; however, this is not a trivial task. A simplified Block diagram of wireless link can be seen in Fig. 1.1a, where a generated baseband signal is fed into a transmitter (Tx) which emits the signal over the air through an antenna by power amplification. This is then picked up by a receiver (Rx), thus recovering the original information emitted over the air. It is fairly easy to see that the block diagram (Fig. 1.1a) is extremely simplified and effectively just provides two boxes that can transmit and receive information. However, the picture does illustrate what this project is trying to discuss, namely how to model such systems through behavioral modelling. This includes defining equations for the individual blocks in the system, generating correct payload in accordance with the specified standard (802.11a), and lots of other efforts. Each step in describing a component introduces a new level of complexity and new decisions to be made however, effectively what is modelled is still the diagram as seen in Fig.1.1a. At a more detailed level, the transmitter (Tx) and receiver (Rx) architectures can be examined as shown in Fig. 1.1b, representing a generic RF transceiver. This abstraction highlights key functional blocks, including the LNA in the receiver, frequency conversion via Up/Down converters driven by a stable local oscillator (LO) signal generated by a frequency synthesizer, and linear power amplification in the transmitter to enable efficient over-the-air transmission. In practice, additional aspects such as signal propagation and modulation also play a significant role in overall system performance which will also be discussed.

In industry, system-level design and verification are often performed using advanced tools such as MATLAB/Simulink, SystemVue, and Cadence AWR, which enable modular construction of complex RF systems. While these tools are powerful, the modular block based approach relies on proprietary implementation details, and removes transparency - This could potentially limit the outcome if not used with sufficient understanding and training in how to use such systems. This project therefore proposes an open-source behavioral modelling framework for RF transceiver design emphasizing easy integration using modularity as seen in commercial tools, while providing full transparency and accessibility, with the final goal of enabling informed system-level analysis. In order to compare the results with actual silicon proven implementations, the parameters used for the individual components through out the system is set in accordance with the following paper [1], and results will be held up with the measured results presented in the paper. Furthermore, this work investigates how system-level specifications translates into design requirements for a Type-II phase-locked loop (PLL) with regards to parameters such as the phase noise, loop-bandwidth and the according performance degradations. All code can be accessed through my personal github page, and the link can be found [here](#), along with extensive documentation that includes simple examples, an explanation of the underlying code architecture, and guidance on effective manipulation of the simulation pipeline. The documentation is available [here](#).



(a) Simplified wireless RF transceiver model.



(b) Generic RF transceiver architecture.

Figure 1.1: Hierarchical representations of RF transceiver systems.

2 |Signal and System Modelling

This chapter introduces the theoretical foundation used for modelling the RF transceiver system. It focuses on complex baseband signal representations, and how these signals are processed in the baseband domain. In addition, key RF impairments are introduced, which are incorporated into the behavioral models of the individual system components.

2.1 Complex Enveloping

A general RF modulated signal can be written on the following form

$$x(t) = R(t) \cdot \cos(2\pi f_c t + \phi(t)). \quad (2.1)$$

It's comprised of an envelope magnitude $R(t)$ controlling the magnitude of the carrier at each time instance t . Besides this, the modulated signal is centered around the passband frequency f_c . This carrier frequency imposes a serious issue when it comes to sampling rate and computational power, since it can be in high orders of GHz and therefore, quickly take up a lot of memory on a given machine. In order to mitigate this for a system analysis, a technique called complex enveloping is used in order to convey the information in the modulation envelope only. Generally the passband signal can be represented as the real valued multiplication of a sinusoidal carrier with the complex envelope seen as

$$s(t) = \text{Re} \{ \tilde{s}(t) e^{j\omega_c t} \}, \quad (2.2)$$

where $\omega_c = 2\pi f_c$. It is known that the passband signal seen in Eq. 2.1 can be written as

$$s(t) = S_I(t) \cos(\omega_c t) - S_Q(t) \sin(\omega_c t), \quad (2.3)$$

where $S_I(t)$ is the in phase component and $S_Q(t)$ is the quadrature component. Given Eulers identity it can be shown that [2]

$$s(t) = \text{Re} \{ (S_I(t) + jS_Q(t)) e^{j\omega_c t} \}, \quad (2.4)$$

and written into to the complex envelope,

$$\tilde{s}(t) = S_I(t) + jS_Q(t). \quad (2.5)$$

2.2 Complex Enveloping in System Modelling

Given the derivation above it was found that the complex envelope model provides a method for converting a passband model down to baseband, by removing the carrier of the passband signal. In a system model this can be exploited in order to dramatically reduce the sampling speed and directly present the signal in a quadrature form (I/Q), meaning all impairments can be inferred directly to each component directly. In order to visualize this, consider the following signal consisting of two tones separated by an offset $\Delta f_1 = f_{\text{offset}1}$, $\Delta f_2 = f_{\text{offset}2}$.

$$s(t) = \cos(2\pi(f_c + \Delta f_1)t) + \cos(2\pi(f_c + \Delta f_2)t). \quad (2.6)$$

The analytic signal corresponding to a real signal $r(t)$ is defined as

$$s_a(t) = s(t) + j\mathcal{H}\{s(t)\}, \quad (2.7)$$

where $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform. Now the analytical signal can be written as

$$s_a(t) = e^{j2\pi(f_c + \Delta f_1)t} + e^{j2\pi(f_c + \Delta f_2)t}. \quad (2.8)$$

Factoring out the carrier term yields

$$s_a(t) = e^{j2\pi f_c t} \left(e^{j2\pi \Delta f_1 t} + e^{j2\pi \Delta f_2 t} \right). \quad (2.9)$$

The term inside the parentheses is the complex envelope.

$$\tilde{s}(t) = e^{j2\pi\Delta f_1 t} + e^{j2\pi\Delta f_2 t}. \quad (2.10)$$

The time domain and frequency domain relation between the real RF signal $s(t)$ and the complex envelope $\tilde{s}(t)$ can be seen in Fig. 2.1.

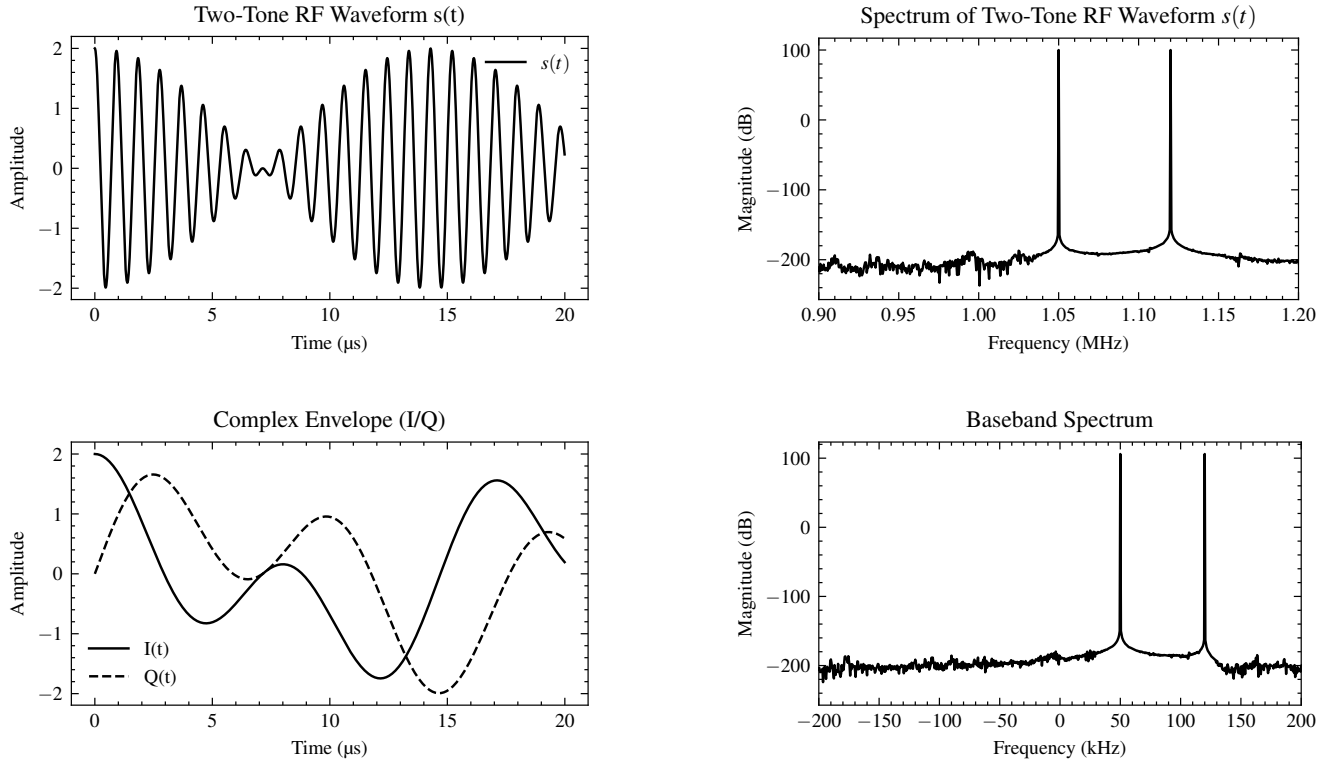


Figure 2.1: Time- and frequency-domain representations of a real RF signal $s(t)$ (top) and its complex envelope $\tilde{s}(t)$ (bottom).

2.3 Non-linear Effects

In classical analog circuit design, active devices such as transistors and vacuum tubes are used as the building blocks for providing functionality such as amplification, mixing and filtering. The behavior of devices such as CMOS and BJT transistors is governed by semiconductor physics and depends strongly on how they are biased. Setting the bias conditions establishes the operating point of the device, which determines how it responds to signals. Accurately describing this behavior requires equations with multiple parameters that capture the device's physical characteristics. In order to simplify this a small-signal approximation is used as a standard abstraction, for sufficiently small input amplitudes, where a non-linear device such as a transistor can be treated as locally linear, greatly simplifying analysis. This assumption holds well in applications where signal amplitudes remain within the linear region of a device's transfer characteristic. In RF systems, however, this condition is not always guaranteed. High signal power levels, the presence of multiple closely spaced carriers, or the summed effect of non-linearity in cascaded stages can drive a device outside its linear operating region resulting in harmonic distortion, gain compression, and intermodulation that finally degrade system performance and must be carefully accounted for in the design process.

In the following sections some of these effects will be introduced and discussed through a polynomial model of a memoryless non-linear system and derive the key figures of merit used to quantify linearity in RF design.

2.3.1 Memoryless System and Harmonic Distortion

A general non-linear memoryless system can be written in its general form as a sum of non-linearities [3]

$$y(t) = \sum_{k=0}^{\infty} \alpha_k x^k(t). \quad (2.11)$$

However, since most circuits are designed to operate mainly in the linear region it's assumed that the effect of higher order terms beyond the third order term are significantly smaller. Therefore, I/O characteristics can be approximated as

$$y(t) \approx \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t). \quad (2.12)$$

Applying a single tone given by $x(t) = A \cos(\omega t)$ to the model yields the following expansion [3]

$$y(t) = \alpha_1 A \cos(\omega t) + \alpha_2 A^2 \cos^2(\omega t) + \alpha_3 A^3 \cos^3(\omega t) \quad (2.13)$$

$$= \frac{\alpha_2 A^2}{2} + \left(\alpha_1 A + \frac{3\alpha_3 A^3}{4} \right) \cos(\omega t) + \frac{\alpha_2 A^2}{2} \cos(2\omega t) + \frac{\alpha_3 A^3}{4} \cos(3\omega t). \quad (2.14)$$

This shows that by applying a single tone, multiple tones will occur at harmonic frequencies at different attenuation rates. This is visualized in Fig. 2.2 for a single tone signal of 1 kHz.

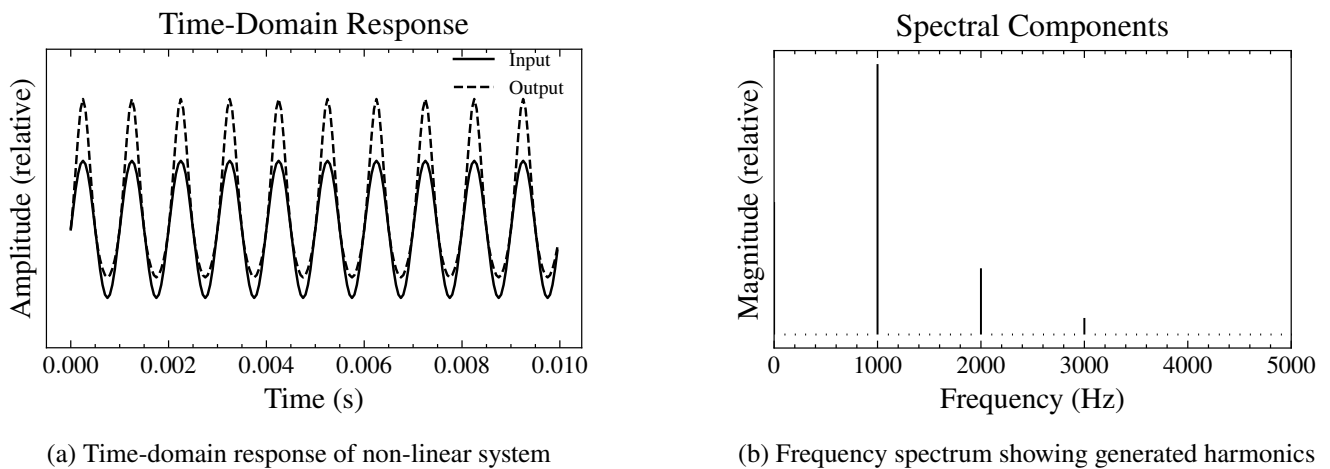


Figure 2.2: Effect of polynomial non-linearity on a single-tone input signal.

A two-tone sinusoidal excitation is now applied to the non-linear system, where the input signal is given by

$$x(t) = A_1 \cdot \cos(\omega_1 t) + A_2 \cdot \cos(\omega_2 t), \quad (2.15)$$

$$y(t) = \alpha_1 (A_1 \cos \omega_1 t + A_2 \cos \omega_2 t) + \alpha_2 (A_1 \cos \omega_1 t + A_2 \cos \omega_2 t)^2 + \alpha_3 (A_1 \cos \omega_1 t + A_2 \cos \omega_2 t)^3. \quad (2.16)$$

This yields a series expansion that, due to mixing of the higher order components at the input of the non-linear system, gives the following output, by discarding the DC terms and the proceeding harmonic terms [3]

$$2\omega_1 \pm \omega_2 : \frac{3\alpha_3 A_1^2 A_2}{4} \cos(2\omega_1 + \omega_2)t + \frac{3\alpha_3 A_1^2 A_2}{4} \cos(2\omega_1 - \omega_2)t, \quad (2.17)$$

$$2\omega_2 \pm \omega_1 : \frac{3\alpha_3 A_2^2 A_1}{4} \cos(2\omega_2 + \omega_1)t + \frac{3\alpha_3 A_2^2 A_1}{4} \cos(2\omega_2 - \omega_1)t. \quad (2.18)$$

The output response for two tones applied to Eq. 2.12 at 900 Hz and 1.1 kHz is shown in Fig. 2.3.

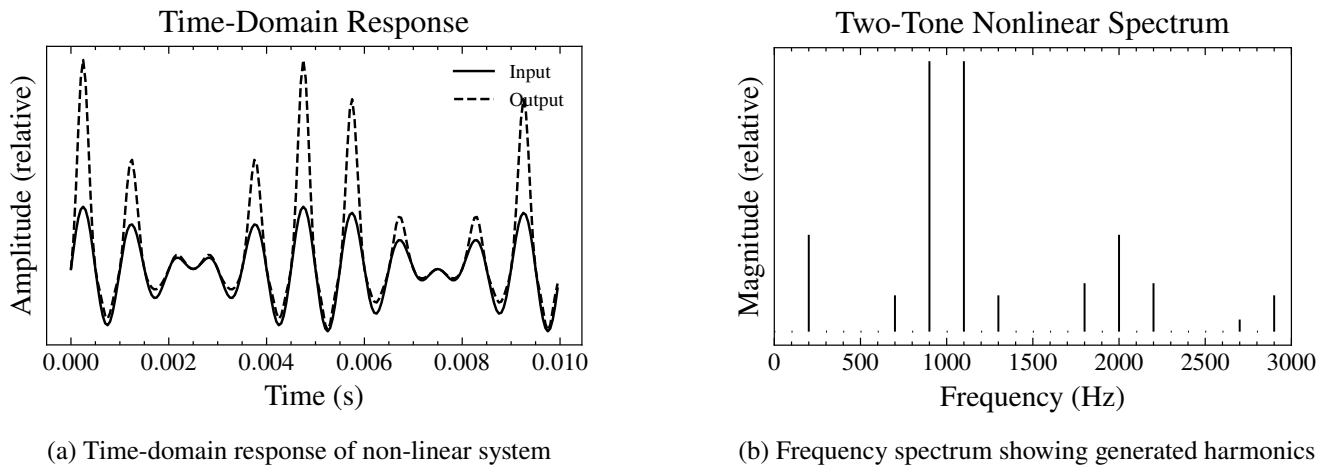


Figure 2.3: Effect of polynomial non-linearity on a two-tone input signal.

As illustrated in Figs. 2.3 and Eq. 2.16, the non-linear system generates higher-order mixing products, in particular the $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$ terms, which give rise to in-band spurs. This is problematic, since these components can interfere with the in-band signal content thus difficult to filter out effectively. The expression can be rewritten in order to showcase the fundamental and second order terms, which yields the following [3]

$$\omega_1, \omega_2 : \left(\alpha_1 A_1 + \frac{3}{4} \alpha_3 A_1^3 + \frac{3}{2} \alpha_3 A_1 A_2^2 \right) \cos(\omega_1 t) + \left(\alpha_1 A_2 + \frac{3}{4} \alpha_3 A_2^3 + \frac{3}{2} \alpha_3 A_2 A_1^2 \right) \cos(\omega_2 t). \quad (2.19)$$

From Eq. 2.19 it's seen that cross modulation appears meaning the amplitude of the carrier signals is impacted by the second interferer tone. This can be shown by rewriting the amplitude of the fundamental ω_1

$$\text{Amplitude}_{\omega_1} = A_1 \left(\alpha + \frac{3}{4} \alpha_3 A_1^2 + \frac{3}{2} \alpha_3 A_2^2 \right). \quad (2.20)$$

Now for $A_2 \gg A_1$ the effective gain can be simplified to [3]

$$G_{eff} = \alpha_1 + \frac{3}{2} \alpha_3 A_2^2. \quad (2.21)$$

Now for $\alpha_1 \alpha_3 < 0$ the effective gain decreases as a function of A_2 . This shows that a non-linear RF system is sensitive to strong interfering signals, which can reduce the gain experienced by the desired tone. The linear gain term is therefore influenced by the interaction between α_1 and α_3 . Depending on their relative sign, the system may exhibit either gain expansion ($\alpha_1 \alpha_3 > 0$) or compression ($\alpha_1 \alpha_3 < 0$). This behavior depends on the device used [3].

2.3.2 Third order Intercept Point

In order to quantify the impact of the non-linear effects - more specifically the intermodulation described in the previous section - the third order intercept point (IIP_3) is defined as the input power at which the extrapolated small-signal curves (i.e., linear curves) of the fundamental and the third order intermodulation (IM3) product becomes equal in magnitude as seen in Fig. 2.4. As established earlier, the IM3 product rises by 3 dB for every 1 dB increase in A_{in} , while the fundamental rises 1 dB per 1 dB. The third order intercept point is found by applying a two tone test similar to what was shown earlier, but in a real scenario the result from such a test is dependent on the actual input amplitude level that was applied to derive the fundamental and the IM3 product outputs. The intermodulation product can from previous discussions be derived as

$$IM = \frac{3}{4}\alpha_3 A_{IIP3}^2. \quad (2.22)$$

Thus the IIP_3 can be found by equating the fundamental and the IM amplitude[3]

$$|\alpha_1 A_{IIP3}| = \left| \frac{3}{4}\alpha_3 A_{IIP3}^3 \right|, \quad (2.23)$$

$$A_{IIP3} = \sqrt{\frac{4}{3} \frac{\alpha_1}{\alpha_3}}. \quad (2.24)$$

In Appendix. A, an analytical evaluation of a two-tone test is carried out, and the beta coefficient is defined based for the complex baseband memoryless cubic expression given as

$$y(t) = \alpha x(t) - \beta x(t)|x(t)|^2 \xrightarrow{IIP3 \text{ relation}} \beta = \frac{\alpha}{2P_{IIP3}}. \quad (2.25)$$

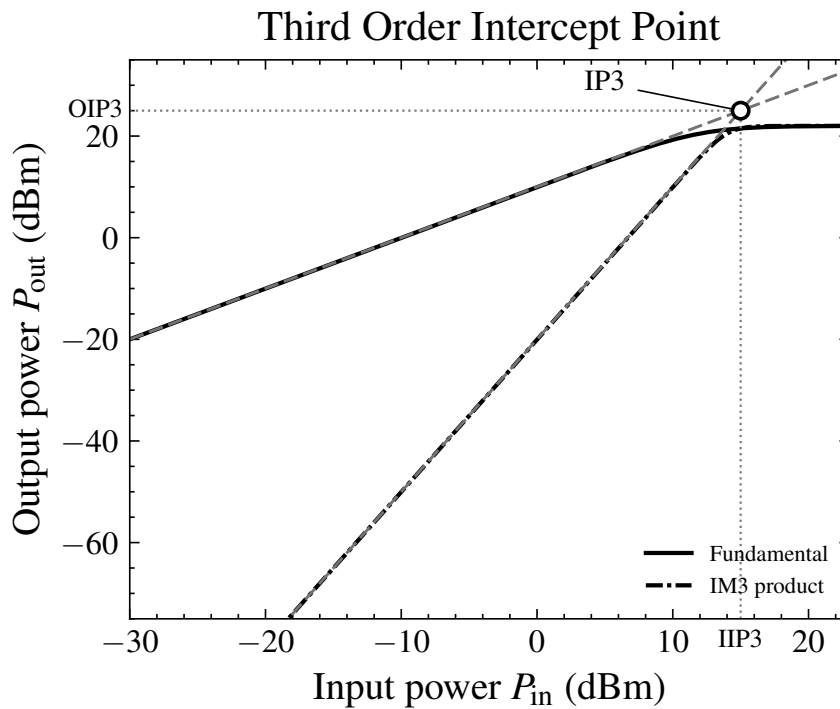


Figure 2.4: Third order intercept point

2.3.3 Compression Point

While intermodulation refers the third order harmonic contribution to a "mixing" product with the fundamental, it was also seen from Eq. 2.21 that compression occurs under specific conditions, namely for $a_1 a_3 < 0$. The compression of a device is by convention defined as the input signal level that causes the output level to drop by 1 dB also known as the 1 dB compression point, which can be seen in Fig. 2.5. The 1 dB compression point can also be referred to power i.e., (P1dB). For calculating the 1 dB compression point the effective gain given by Eq. 2.21 is set equal to the ideal small signal gain minus 1 dB [3].

$$20 \log |\alpha_1 + \frac{3}{4} \alpha_3 A_{in,1dB}^2| = 20 \log |\alpha_1| - 1 \text{ dB}. \quad (2.26)$$

Now the input amplitude can be written as

$$A_{in,1dB} \approx \sqrt{0.145 \left| \frac{\alpha_1}{\alpha_3} \right|}. \quad (2.27)$$

Now this could be converted to power domain for finding P1dB, since $P = \frac{A_{peak}^2}{2R}$. Note that Eq. 2.27 is not an exact result since it depends on the expansion done in Eq. 2.19. In Appendix. B a derivation for the beta factor in the complex baseband memoryless cubic expression was found with regards to the 1 dB compression point, and is given as

$$\beta = (1 - c) c^2 \frac{\alpha^3}{P_{out,1dB}}. \quad (2.28)$$

where $c = 10^{-1/20}$.

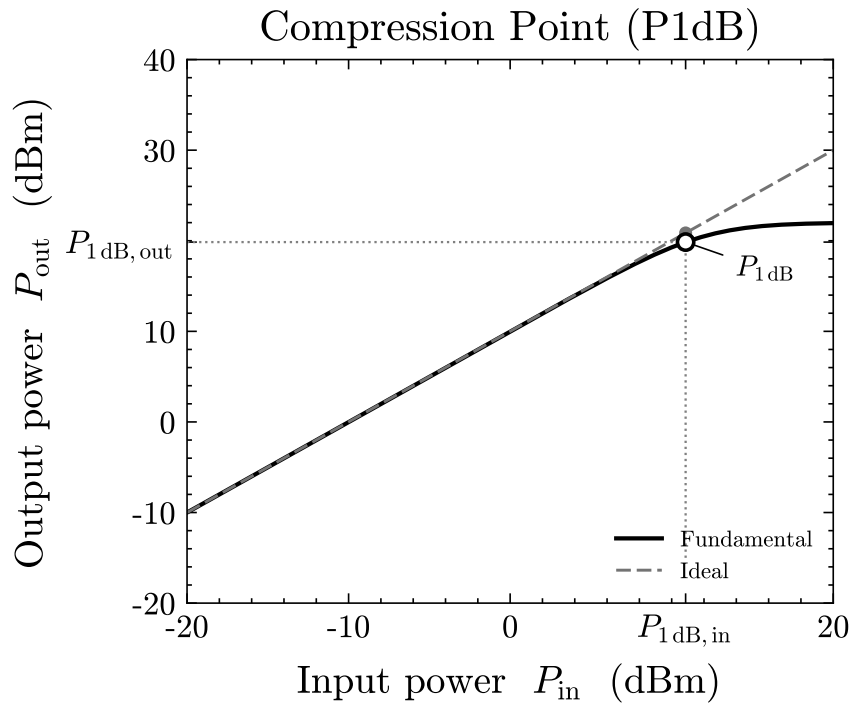


Figure 2.5: Compression point: P1dB

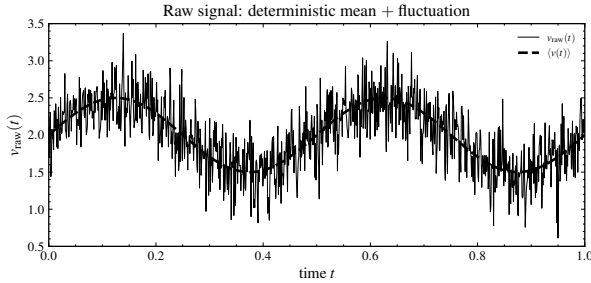
2.4 Noise In RF Systems

In the following section noise in RF systems will be discussed with regards to a theoretical description of how to understand noise modelling and general device/cascaded noise.

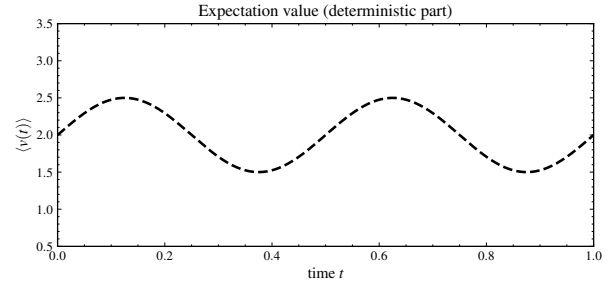
2.4.1 Noise Spectrum

The concept of noise is a rather abstract phenomena that could be broken into several layers however, at the core noise simply comes from electronics deviating from a perfect synchronization pattern in a given conductor thus resulting in random fluctuations. While this is a quite extreme oversimplification, it does emphasize the main concept of how noise is modelled, namely as a stochastic process. Imagine a time dependent stochastic voltage $v_{raw}(t)$ as seen in Fig. 2.6a. The long term average (expectation value $\langle v(t) \rangle$) represents a DC averaging over the entire signal length, and thus subtracting this from $v_{raw}(t)$ yields a zero-mean quantity $v(t)$ as seen in Fig. 2.6b, 2.6c. This quantity could be visualized as an additive noise source imposed to a given output signal of an arbitrary amplifier [4]. At this point it might seem impossible to capture the behavior of noise given this instantaneous complexity in the time-domain however, given that noise can be considered as a stationary process where the steady state auto correlation (ACF) can be written as $C(\tau) = \langle v(\tau)v(0) \rangle$ allows the usage of the Wiener-Khintchine theorem to apply the Fourier Transform directly to the ACF in order to derive the power spectral density (PSD) i.e., the distribution of power across multiple frequencies. The frequency integral over the one sided PSD ($S_{v,v}(\omega)$) as presented in [4], is given by Eq. 2.29, and it allows complex time-domain fluctuations to be interpreted as "simple" statistical variance and power spectral density as seen in Fig. 2.6d - Note that the fourier transform used to derive Eq. 2.29 is in fact the one-sided PSD [4]. In most practical cases the combined effect of noise contributions approaches a Gaussian distribution which allows noise to be modelled by the variance of a stochastic distribution given by

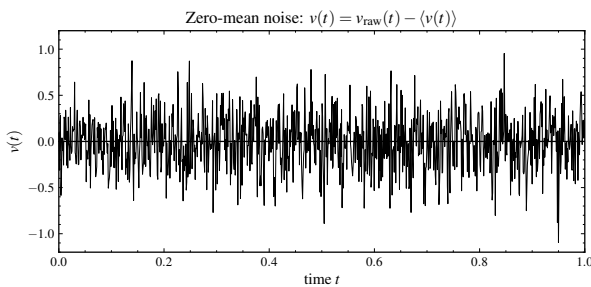
$$\sigma^2 = \int_0^\infty \frac{d\omega}{2\pi} S_{v,v}(\omega). \quad (2.29)$$



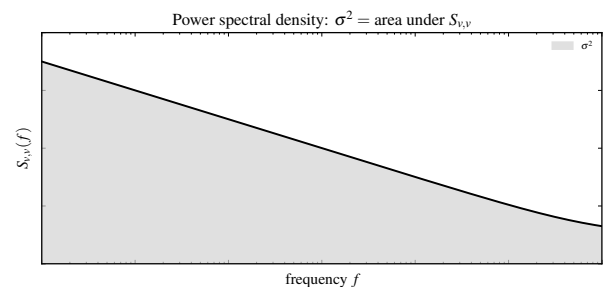
(a) Raw signal $v_{raw}(t)$.



(b) expectation average $\langle v(t) \rangle$.



(c) Zero-mean noise component.



(d) Power spectral density plot.

2.4.2 Device Noise

One of the most fundamental contributions to the overall noise spectrum of a device is the thermal noise, caused by temperature dependent random motion of electronics in a given conductor [4] [3]. This noise contribution can be modelled by a simple voltage source with a PSD given by Eq. 2.30 where k is Boltzmanns constant, R is the effective resistance, T is the temperature in Kelvin and B is the bandwidth.

$$\overline{V_n^2} = 4kTRB. \quad (2.30)$$

The equation above is also referred to as the Johnson-Nyquist Noise and allows to estimate thermal noise contribution in a given conductor medium, for instance the thermal noise in saturation biased MOS transistors is approximated by a current source tied between the source and drain terminals given by $\overline{I_n^2} = 4kT\gamma g_m$ where γ is the excess noise coefficient and g_m is the transconductance, while for a BJT this could be modelled by the thermal noise from the base resistance [3]. Another large contributor to overall device noise in modern semiconductor technology is also called flicker ($1/f$) noise. This is a low-frequency phenomena stemming from the internal physics of semiconductor materials which will not be covered in greater detail. Given flicker noise low-frequency characteristics this is not a crucial phenomena in practically linear devices such as amplifiers etc. however, for non-linear devices such as oscillators this proves to be quite problematic since non-linear mixing processes effectively up-converts the low noise fluctuations into the RF domain. This can be reduced by using phased locked looping i.e., a closed-loop bandwidth to suppress these contributions [4].

2.4.3 Cascaded Noise Stages

In a large system an evaluation of the overall noise figure of a given chain composed by multiple elements is essential to derive the overall noise factor (i.e., $F = \frac{SNR_{in}}{SNR_{out}}$). This is done using Eq. 2.31 also called Friis formula for cascaded noise.

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots + \frac{F_n - 1}{G_1 G_2 \dots G_{n-1}}. \quad (2.31)$$

As seen from the equation above this shows that the overall noise performance of a given cascaded system as seen in Fig. 2.7 is largely determined by the amplification of the first stage(s) [3].

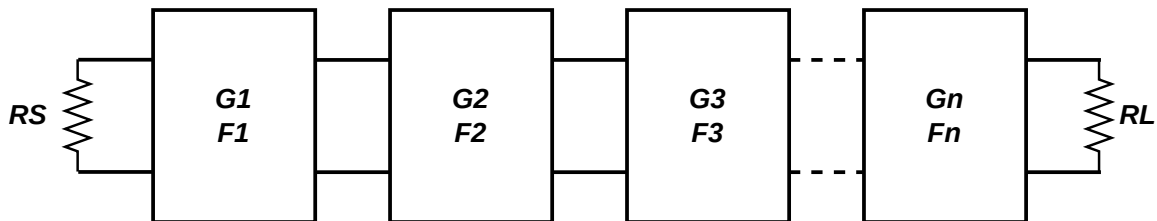


Figure 2.7: Cascaded Stages

3 | Behavioral Modelling

In the following section the components used in the system model will be described with regards to the underlying theory of each element and the actual implementation with essential code snippets. Each block described in the following chapter will follow the general block diagram as seen in Fig. 1.1b with the possibility of further illustrative expansion.

3.1 Signal Generation

At the first stage of the system lies the digital signal processor, in which the input signal for the system will be generated and recovered accordingly. In the following section the elaborated block diagram seen in Fig. 3.1 will be explained with regards to the implementation in the actual code. As seen in the block diagram this follows a three step process namely the Pseudorandom-Bit-Sequence generation (PRBS), quadrature-amplitude-modulation (QAM) and the Orthogonal-Frequency-Division-Multiplexing (OFDM). The signal is generated with respect to the IEEE 802.11a standard [5]. Please refer to Appendix. C to see a small note on the IEEE specifications.

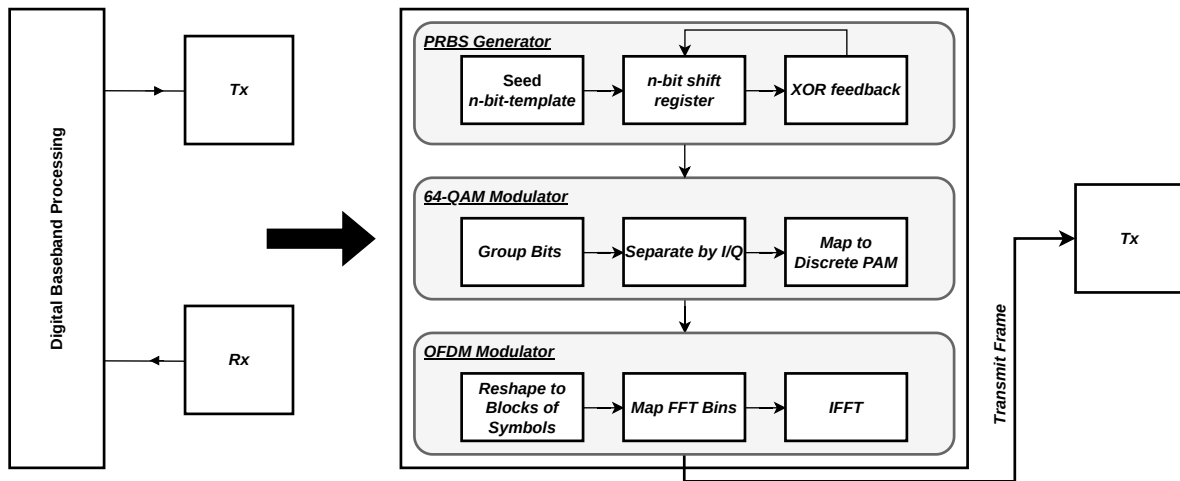


Figure 3.1: Expanded block diagram of the input signal generation chain for a complex OFDM system, assuming the receiver performs the inverse operations for signal recovery.

3.1.1 Baseline Calculation

At first the baseband input signal is defined by a QAM modulation index, i.e., $M = 64$, for a total of 6 bits per symbol. The channel bandwidth of the signal is specified in the 802.11a PHY standard [5] and given as $BW = 20$ MHz. Here the FFT size is set as $n_fft = 64$, which ultimately determines the time-domain symbol length together with the cyclic prefix (in this case set to $cp_len = 16$), giving a total symbol length of $64 + 16 = 80$ samples. From here 52 bins are allocated for useful data, and the rest is used for DC nulling and guard bands, as specified in 802.11a. The total number of OFDM symbols to be transmitted sequentially is given as $num_ofdm_symbols = 64$, resulting in a total of 3328 transmitted QAM symbols, and the total amount of bits is then given as $B_{total} = 19\,968 \approx 20\,000$ bits per frame. Now each frame will have a duration time to resolve each symbol given as

$$T_{symbol} = 80 \text{ samples} \cdot \frac{1}{20 \text{ MHz}} = 4 \mu s, \quad (3.1)$$

$$T_{frame} = N_{sym} \cdot T_{symbol} = 64 \cdot 4 \mu s = 256 \mu s, \quad (3.2)$$

resulting in a data rate given by

$$R = \frac{B_{\text{total}}}{T_{\text{frame}}} = \frac{19\,968 \text{ bits}}{256 \mu\text{s}} \approx 78 \text{ Mbit/s.} \quad (3.3)$$

Note this is above the specified 54 Mbit/s as specified in 802.11a, however in this model service bits, tail bits etc.. is not accounted for.

3.1.2 Pseudorandom Generation

When verifying the behavior of communication channels it's important to ensure randomization in the test procedure to be certain that the transmitter, in this case, is not working in extreme edges of operation. This could be in the case where the input string bits have long repetitions of 1's and 0's thus stressing or slacking the hardware chain, and not capturing the true holistic behavior of the TRx. Therefore a pseudorandom generator was made in order to scramble the generated input bits accordingly. The bit source is implemented as a Linear Feedback Shift Register (LFSR) based PRBS generator, supporting PRBS-7 and PRBS-15 sequences. The generator is initialized with a given seed, which essentially is a bit template with a length given by the order of the bit sequence. As the seed is initialized into the generator, it is sifted each clock cycle, while feeding back to the XOR of two taps defined by the sequence polynomial. For PRBS-15 the polynomial is $x^{15} + x^{14} + 1$, giving a maximal-length sequence of $2^{15} - 1$ bits before repeating. This generates a deterministic but statistically uniform binary sequence used for the modulator [6]. A code of the object generation is seen in the following snippet.

Listing 3.1: PRBS Object Generation.

```

1 dummy_sig = Signal(x=np.array([], dtype=np.uint8), fs_hz=fs_hz, meta={"name": "
  PRBS15_Input_Bits"})
2
3 prbs_gen = PRBSBitSource(
4     name="tx_bits",
5     params=PRBSParams(
6         order=15,
7         n_bits=n_qam_symbols_transmitted * bits_per_symbol,
8         seed=123,
9     ),
10 )
11
12 sig_bits = prbs_gen.process(dummy_sig)
13 bits = sig_bits.x

```

As it can be seen from above a dummy signal object is made, as given the structure explained earlier. At this point this is just a frame that will contain the information generated by PRBSBitSource. The PRBS generator initializes n_{bits} , given by the total amount of bits.

3.1.3 QAM and OFDM Symbol Generation

Using the random sequence the bits can now be allocated onto QAM symbols and grouped onto OFDM carriers. The QAM modulator groups incoming bits into chunks of $\log_2(M) = 6$ and splits each chunk into I and Q terms and maps each part to an odd PAM level $(-7, -5, \dots, +7)$, and combines them into a complex symbol $I + jQ$. The Power is then normalized to unit average i.e., $\mathbb{E}\{|x(t)|^2\} = 1$. Furthermore an optional flag for Grey-Coding is available, in order to make sure that the bit-to-symbol mapping is reordered such that adjacent constellation points differ by only 1 bit. This is critical since landing on neighbors with only single bit flips minimizes the bit error rate and can be seen in Fig. 3.2. A pseudo code of the implementation can be seen below.

Listing 3.2: QAM modulation.

```

1 for each group of log2(M) bits:
2   split bits -> I-bits | Q-bits
3   map I-bits -> odd PAM level in [-7 ... +7]
4   map Q-bits -> odd PAM level in [-7 ... +7]
5   symbol = I_level + j * Q_level
6
7 normalize all symbols to unit average power

```

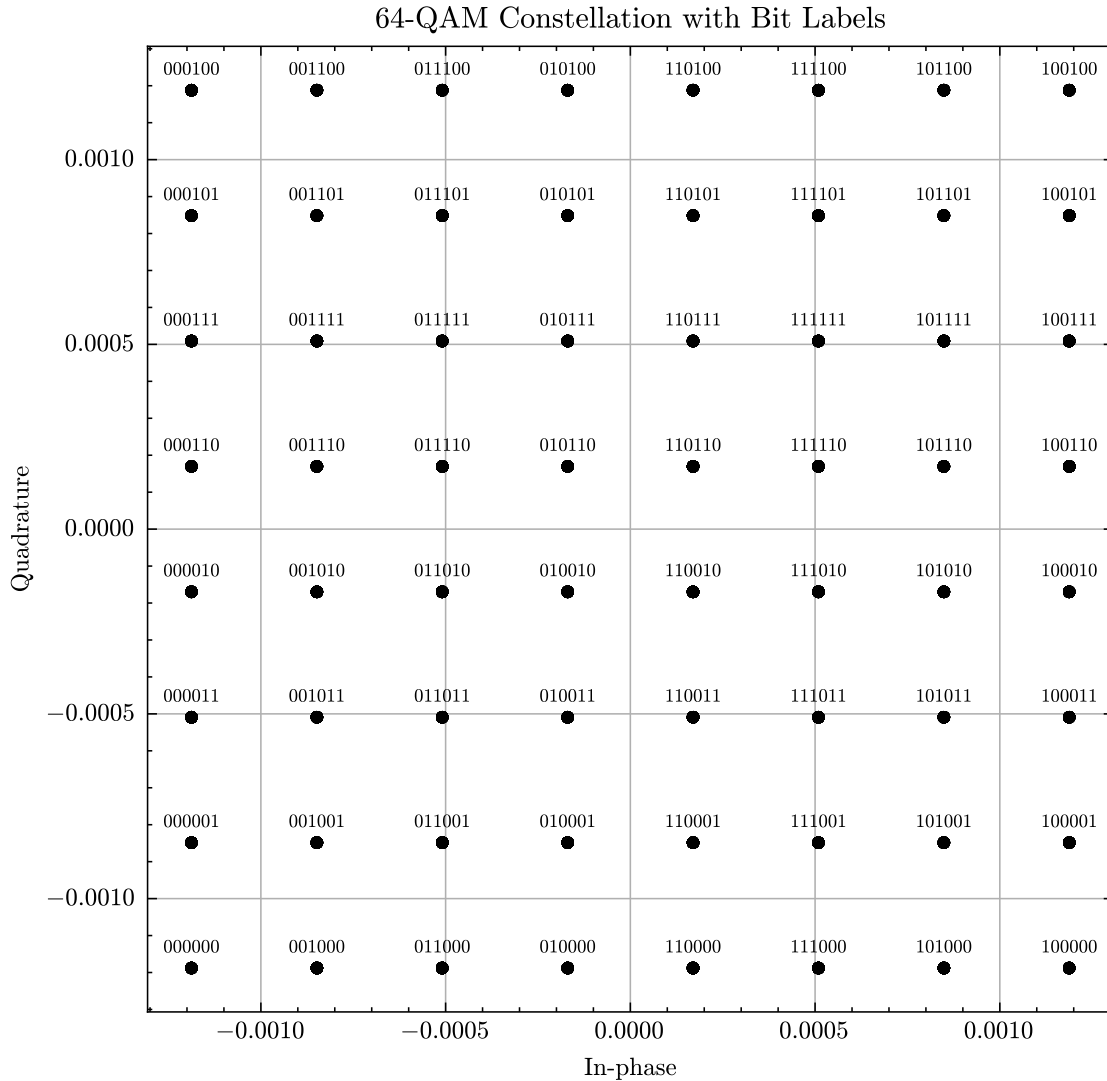


Figure 3.2: Constellation Points of Generated Input Signal.

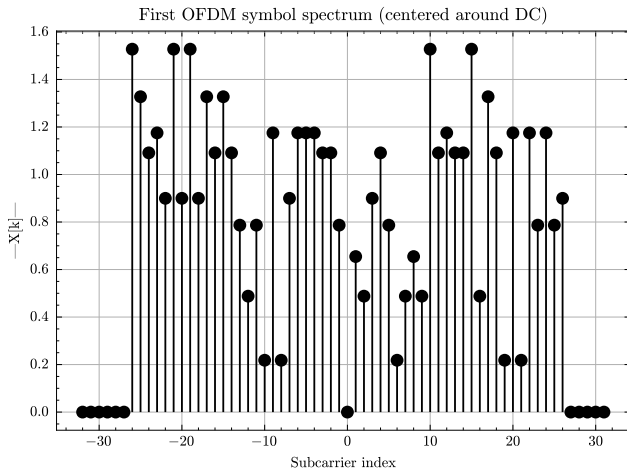
The next step is for the OFDM modulator to place the QAM symbols onto 52 active FFT bins (split symmetrically around DC which is left null). Afterwards an IFFT is executed to produce the time-domain symbol, and prepends the cyclic prefix (last 16 samples copied to front). This can be summarized in 4 steps. First a flat array of QAM symbols (in complex form) is reshaped into blocks of 52 symbols, which essentially is the sub-carriers.

$$\mathbf{d}_m = [d_{m,0}, d_{m,1}, \dots, d_{m,51}], \quad m = 0, 1, \dots, 63. \quad (3.4)$$

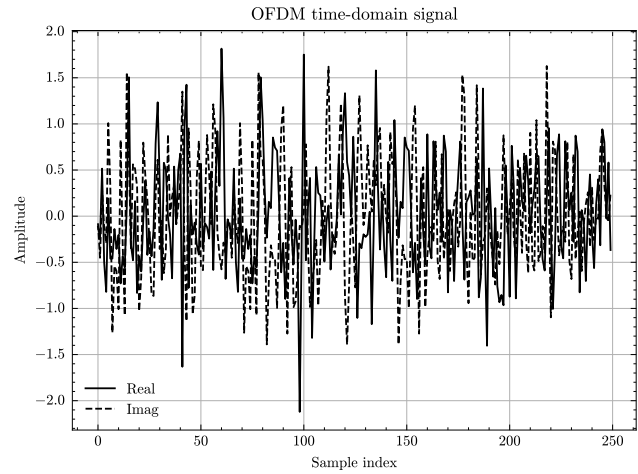
Where each $d_{m,k}$ is a 64-QAM symbol. Each block is mapped onto an $N_{\text{FFT}} = 64$ point frequency vector $X_m[k]$, with 52 data symbols placed symmetrically around DC, as seen in Fig. 3.3a. The time-domain OFDM symbol is produced by the N_{FFT} -point IFFT.

$$x_m[n] = \frac{1}{N_{\text{FFT}}} \sum_{k=0}^{N_{\text{FFT}}-1} X_m[k], e^{j2\pi \frac{kn}{N_{\text{FFT}}}}. \quad (3.5)$$

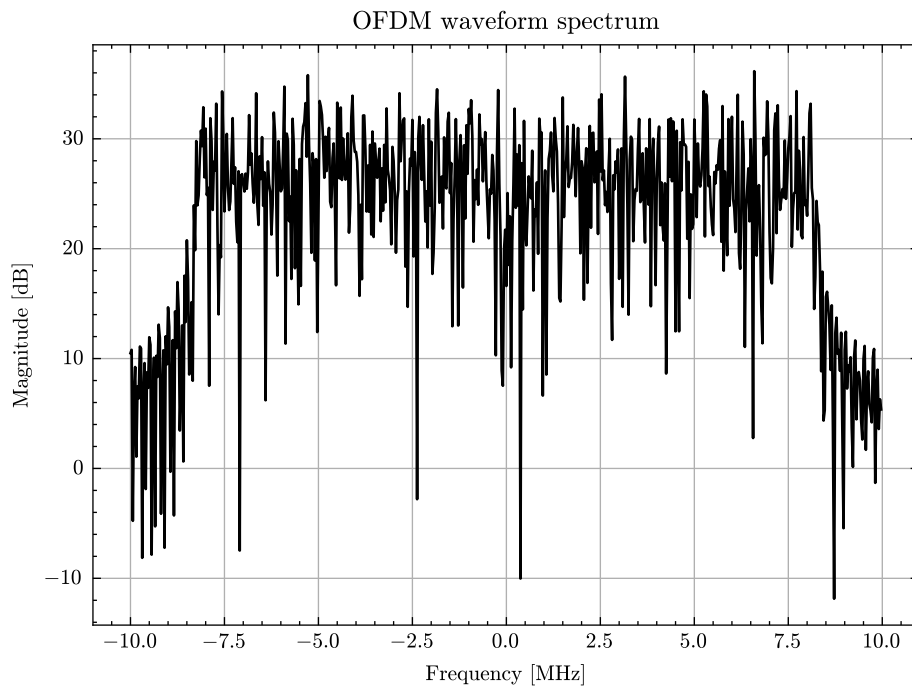
The last $N_{CP} = 16$ samples of $x_m[n]$ are prepended, and the $N_{sym} = 64$ symbols are concatenated into the final transmit frame $N_{sym}(N_{FFT} + N_{CP})$. Below the process is illustrated, where Fig. 3.3a illustrates the discrete sub-carrier allocation, and Fig. 3.3b illustrates the time domain signal after inverse fourier transformation, and Fig. 3.3c shows the waveform spectrum. In order to control the input power level of the samples the generated samples are normalized to a given dBm by simply providing a scaling factor to the samples.



(a) OFDM frequency bins (subcarrier allocation).



(b) Time-domain OFDM signal after IFFT and cyclic prefix.



(c) Frequency spectrum of the transmitted OFDM frame.

Figure 3.3: OFDM signal at key stages of the modulation chain.

3.2 Mixing and Frequency Synthesis

In order to transmit the signal into a selected frequency band, the signal must be up-converted to RF domain. This is done using mixers, and frequency synthesizers like Phased locked loops (PLLs), or simple voltage controlled oscillators (VCO). In the same way as up-conversion is needed, down-conversion performs the inverse operation and thus brings a given RF signal down to baseband. These conversion schemes are known as direct conversion architectures, and a block diagram elaborating the implementation seen in the generic transceiver Fig. 1.1b, can be seen in Fig. 3.4. Note that the outputs from the generated input signal already is separated into in phase and quadrature (I/Q) form, meaning the data is structured to be 90° out of phase.

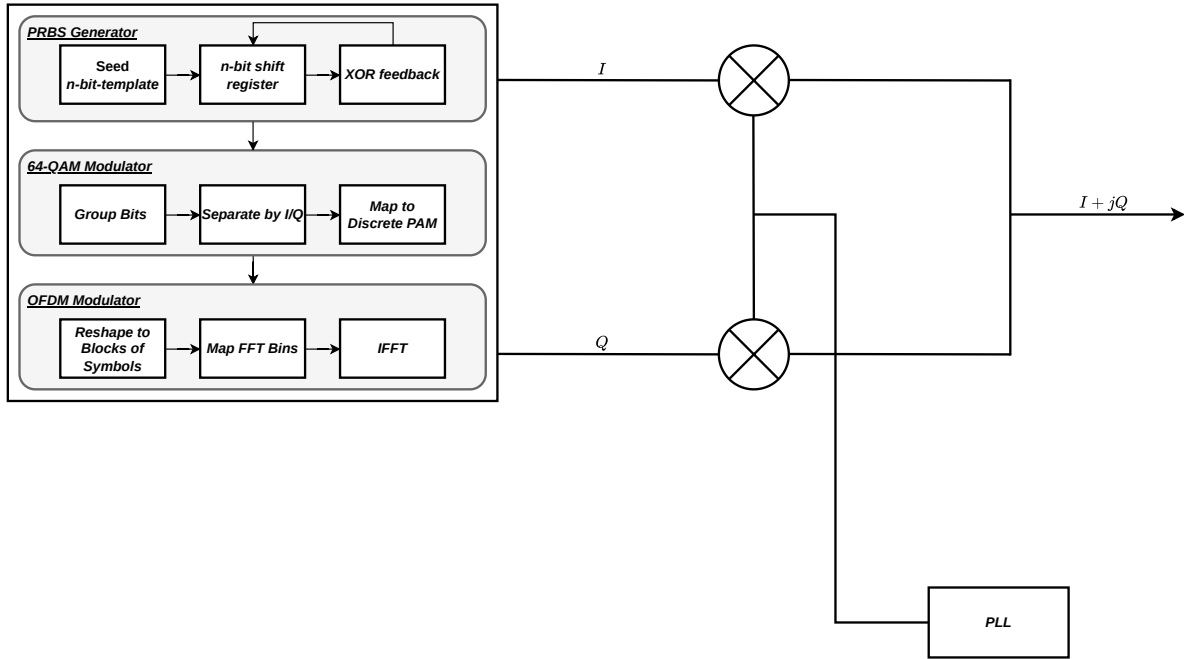


Figure 3.4: Expanded block diagram of the up-conversion link.

3.2.1 Mixing

Designing mixers in modern technologies, proves to be challenging given the highly non-linear behavior mixers exhibits. In the ideal world a mixer would simply provide a functional multiplication of two input signals seen as

$$y(t) = x(t)\cos(2\pi f_{LO}t), \quad (3.6)$$

where $x(t)$ is the baseband signal. Using simple trigonometric expansion it's seen that the output terms are centered around $f_{LO} \pm f_{BB}$, however as seen in Sec. 2.3 due to non-linear behavior, intermodulation occurs. In general, a mixer should compromise intermodulation and noise performance, since IP_3 scales inversely with input gain, while the noise performance is improved when increasing the gain (Rx input referred) [3]. While the third order intermodulation products (IM3) proved to be problematic due to in band spurs, the second order intermodulation product also needs to be carefully considered. This is especially important in direct down conversion structures since the second order term will produce in band terms at $\omega_1 - \omega_2$ similar to what is trying to be achieved directly with the LO. Imagine a radio trying to listen to a signal at 900 MHz, this would be achieved with an ideal mixer providing a DC centered output, however since perfect isolation of the channel cant be ensured, nearby signals may intermodulate in the mixer. Assume two signals at 920 MHz and 921 MHz is seen at the input of the mixer, then the second order intermodulation will produce spurs at $f_1 - f_2 = 1$ MHz. The non-linear effects of the mixer is modelled using the complex baseband model as shown in Appendix A thus the non-linear coefficients are found as

$$\gamma = \frac{\alpha}{\sqrt{2P_{IIP2}}}, \quad (3.7)$$

$$\beta = \frac{\alpha}{2P_{IIP3}}. \quad (3.8)$$

This yields the non-linear cubic transfer function given as

$$y(t) = \alpha x(t) + \gamma \cdot x(t) \cdot |x(t)| - \beta \cdot x(t) \cdot |x(t)|^2. \quad (3.9)$$

Besides the non-linear effects, the noise contribution of the mixer can be modelled using its noise figure. This is expressed as an equivalent input-referred noise source with gain scaling to the output [7].

$$P_n = (F - 1) \cdot kT \cdot B \cdot G. \quad (3.10)$$

From here the power is split evenly between real and imaginary part for complex baseband modelling and the variance of a Complex Circularly-Symmetric Gaussian noise source is given as.

$$\sigma^2 = \frac{P_n}{2}. \quad (3.11)$$

This allows to model a white noise source with a total average power of P_n [8]. A pseudo code of the mixer implementation can be seen in Listing 3.3.

Listing 3.3: Pseudo-code of RF Mixer Block with Nonlinearity and Noise Modeling

```

1 function MixerBlock.process(signal x)
2
3     // Linear gain
4     G      = 10^(gain_db / 10)
5     alpha  = sqrt(G)                // voltage gain
6
7     // Third-order nonlinearity (IIP3 model)
8     beta   = alpha / (2 * P_IIP3)
9     y      = alpha * x - beta * |x|^2 * x
10
11    // Second-order nonlinearity (optional IIP2 model)
12    if iip2_enabled:
13        a2 = alpha / sqrt(2 * P_IIP2)
14        y  = y + a2 * x^2
15
16    // Thermal noise (input-referred, output-scaled)
17    P_noise = (F - 1) * k * T * G * (fs / 2)
18
19    sigma   = sqrt(P_noise / 2)
20
21    n_real  = sigma * N(0,1)
22    n_imag  = sigma * N(0,1)
23    n       = n_real + j * n_imag
24
25    y = y + n
26
27    return y

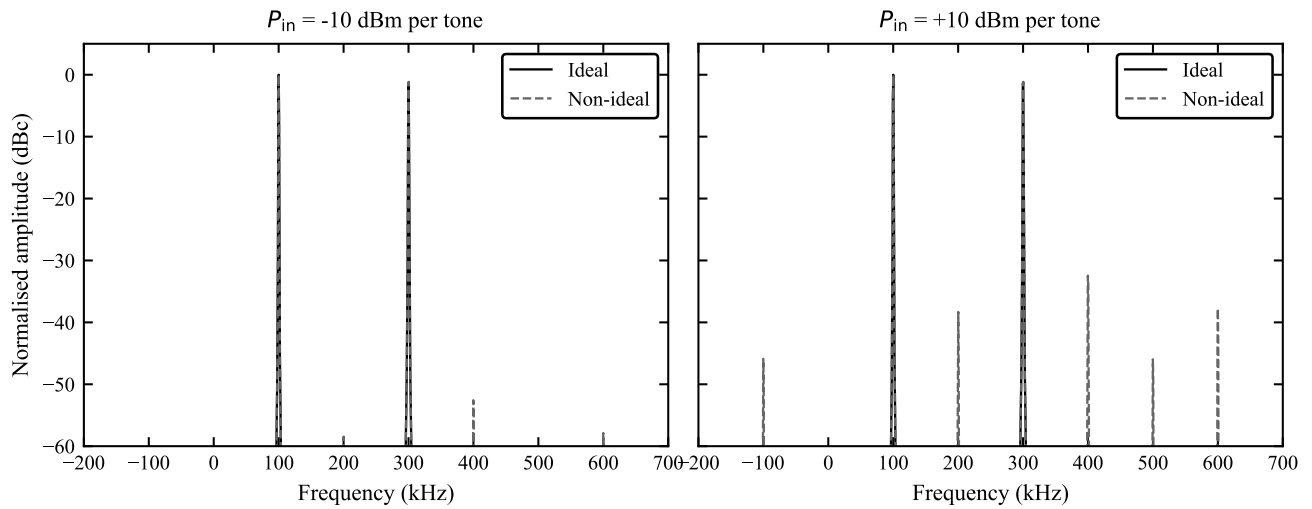
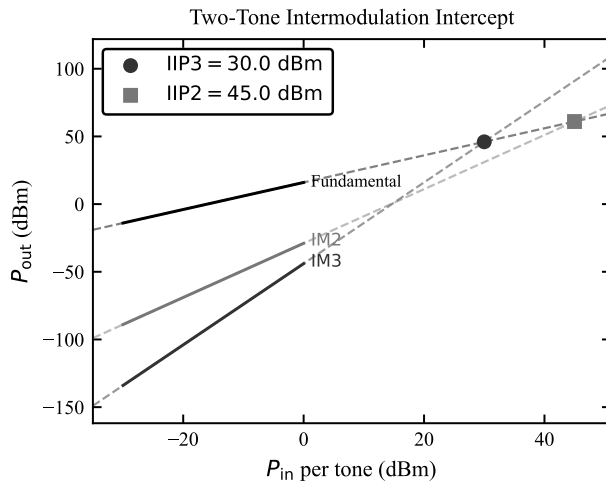
```

The mixer is verified using a two-tone complex-baseband test signal consisting of tones at $f_1 = 100$ kHz and $f_2 = 300$ kHz, sampled at $f_s = 10$ MHz, and the specifications for the mixer is given in Tab. 3.1

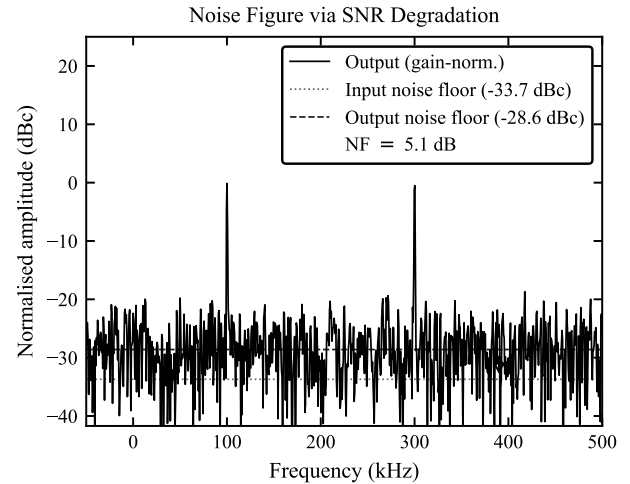
The response of the mixer is seen in Fig. 3.5a with respect to the ideal response. This is plotted at two different input power levels to emphasize the impact of the non-linear terms as the input power increases. As seen the functional mixer produces IM2 terms: $2f_1$, $2f_2$, $f_1 + f_2$ and IM3 terms: $2f_1 - f_2$, $2f_2 - f_1$. In Fig. 3.5b an extrapolation of the second and third order intercept point from the two tone test can be seen, while Fig. 3.5c shows the noise figure of the mixer given by the ratio between the input and output noise floor.

Table 3.1: Mixer and PLL model parameters.

Parameter	Symbol	Value	Unit
<i>Mixer</i>			
Conversion gain	G	10	dB
Noise figure	NF	5	dB
Input 3rd-order intercept	IIP3	30	dBm
Input 2nd-order intercept	IIP2	45	dBm

(a) Output relationship between ideal and non-ideal mixer for -10 dBm and 10 dBm input.

(b) Extrapolation of the second and third order intercept points.



(c) Output spectrum and noise figure evaluation.

Figure 3.5: Functional verification of the mixer model using a two-tone test.

3.2.2 Phased Locked Loop (PLL)

The phased locked loop is an essential component in applications where accurate frequency synthesis is needed. In the application as seen in Fig. 3.4 the PLL is responsible for generating the local oscillator (LO) used for up-conversion (Tx) and down-conversion (Rx). During the conversion the modulated signal of interest will be introduced to undesired elements given the non-linearity in both the mixer and PLL. One of the most critical parameters that has the biggest influence on the transmitted and received signal is the phase noise of the LO signal, generated by the PLL, since this will cause phase rotation that translates directly to errors in the constellation diagram. To understand how phase noise is modelled it is essential to understand how phase noise can be interpreted. A PLL can be build up in various different ways, where some of the most common PLL's - also known as type II PLL - are charge-pump based as seen i Fig. 3.6. The charge-pump is responsible for delivering the control voltage to the (VCO) through a loop filter. Along with the VCO itself the charge-pump can introduce significant phase noise.

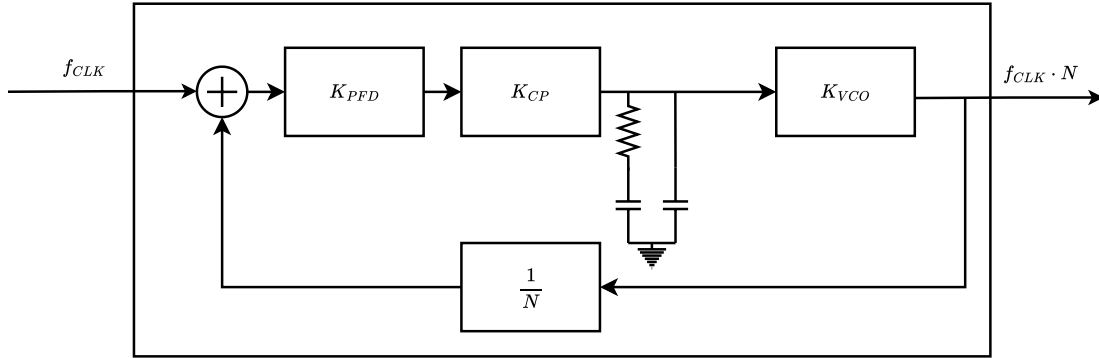


Figure 3.6: Block diagram of generic Type II Phased Locked Loop.

3.2.3 Understanding Phase Noise

To model the stochastic process underlying phase noise, the *Wiener process* (Brownian motion) is employed. The Wiener process describes the phase of a free-running oscillator driven by white noise sources, and is governed by the stochastic differential equation [9]

$$\frac{d}{dt}\theta(t) = F(t), \quad (3.12)$$

where $F(t)$ is white Gaussian noise. Since integration of white noise yields unbound phase variance the PSD of a free running oscillator is given as

$$S_{\theta}(f) = \frac{S_F(f)}{(2\pi f)^2}. \quad (3.13)$$

This shows that the Wiener process only captures $\frac{1}{f^2}$ noise (Lorentzian slope) and will yield an unbounded PSD since $S_{\theta}(f) \propto \frac{1}{f^2}$ thus integrating over the PSD will make the spectrum diverge at $f = 0$. However this is only applicable for a free running VCO. In reality the phase noise spectrum of an PLL is affected by flicker noise, loop suppression of low-frequency phase noise etc. As far as the VCO is concerned the closed-loop PLL spectrum will appear as a high pass filtered free running VCO, thus Brownian motion with a restoring term (effectively replacing pure integrator with finite DC gain) proportional to the loop bandwidth $\omega_L = 2\pi f_L$ [9] (first order high pass filter).

$$\frac{d}{dt}\theta(t) + \omega_L\theta(t) = F(t). \quad (3.14)$$

Through elaborate mathematics as shown in [9] it is seen that a steady state power spectral density of the Ornstein-Uhlenbeck can be obtained from the Wiener-Khinchin theorem, yielding

$$S_{\theta}^{\text{PLL}}(f) = \frac{D_{\phi}/\pi^2}{f^2 + f_L^2}, \quad (3.15)$$

Which is a Lorentzian spectrum, where D_{ϕ} is the phase diffusivity i.e., the slope of the phase variance as a function of time.

3.2.4 Lorentzian Phase Noise Model

Given the conclusion from before it is seen that the PLL phase noise can be modelled as a Lorentzian spectrum, and the total phase jitter can be obtained by integrating the reference noise floor S_{LF} and the VCO noise over the full frequency range. Following [10], the VCO phase noise is expressed as $S_{VCO}(f) = \alpha/f^2$, where α is an amplitude scaling constant. The total phase variance for a first-order PLL model is given by

$$\sigma_{\phi}^2 = 2 \int_0^{\infty} \left[S_{LF} \frac{1}{1 + (f/f_L)^2} + \frac{\alpha}{f^2} \frac{(f/f_L)^2}{1 + (f/f_L)^2} \right] df. \quad (3.16)$$

Evaluating the integral gives

$$\sigma_{\phi}^2 = \pi S_{LF} f_L + \pi \frac{\alpha}{f_L}. \quad (3.17)$$

This shows that the PLL is a combination of noise contributions when $f \ll f_L$, i.e., noise contributions in the loop (PFD, CP, reference etc.), and noise for $f \gg f_L$ i.e., free running VCO $1/f^2$ noise essentially leaking through. To model this a behavioral block for the PLL is implemented using the first order Lorentzian closed-loop as seen in Eq. 3.16. The PSD is calculated by and embedded function employing the closed-loop transfer function as seen in Listing 3.4

Listing 3.4: PLL PSD.

```

1 def get_psd(self, f: np.ndarray) -> np.ndarray:
2     f_L = self.p.f_L
3
4     # Avoid division by zero at f=0
5     f_safe = np.where(f == 0, np.finfo(float).eps, f)
6
7     lp_factor = 1 / (1 + (f_safe / f_L)**2) # low-pass: reference noise
8     hp_factor = (f_safe / f_L)**2 / (1 + (f_safe / f_L)**2) # high-pass: VCO
9     noise
10
11     S_phi = self.SLF * lp_factor + (self.alpha / f_safe**2) * hp_factor
12
13     if self.p.enable_ofdm_weighting:
14         denom = (np.pi * f_safe * self.p.Tu)**2
15         h_bbf = 1 - np.divide(
16             np.sin(np.pi * f_safe * self.p.Tu)**2,
17             denom,
18             out=np.zeros_like(f_safe),
19             where=denom != 0
20         )
21         S_phi *= h_bbf
22
23     return S_phi

```

In order to convert the frequency domain PN into time domain noise, a new function called `generate_lo_impairment` has been made. This function starts by creating a one-sided frequency axis constructed with resolution

$$\Delta f = \frac{f_s}{N}, \quad (3.18)$$

covering offsets from DC up to the Nyquist frequency $f_s/2$. The Lorentzian PSD $S_{\phi}(f)$ is evaluated at each bin using `get_psd`. To ensure the resulting time-domain sequence has the correct variance, the amplitude of each bin must satisfy Parseval's theorem (Time domain and Frequency domain energy must match). Using Numpy ifft library utilizes the following form of the IFFT and FFT,

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad (3.19)$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}. \quad (3.20)$$

Now using `rfft` gives positive frequencies only i.e., the one sided spectrum. The ifft of the phase spectrum can be written as

$$\phi_t[n] = \frac{2}{N} \cdot \sum_{k=0}^{N-1} |\phi_f[k]|^2, \quad (3.21)$$

and therefore the mean power is given as

$$\overline{\phi^2} = \frac{2}{N^2} \sum_k |\phi_f[k]|^2, \quad (3.22)$$

which must equal $\sum_k S_\phi(f_k) \Delta f$ since the integral of the PSD gives the total variance of the signal (i.e., average power). This requires

$$\mathbb{E}[|\phi_f[k]|^2] = S_\phi(f_k) \Delta f \cdot \frac{N^2}{2}. \quad (3.23)$$

Since $\mathbb{E}[|\phi[k]|^2] = 2$ for a normalization where $E[x^2] = 1$ and $E[y^2] = 1$, the correct scaling factor is

$$\phi_f[k] = \sqrt{S_\phi(f_k) \Delta f} \cdot \frac{N}{2}. \quad (3.24)$$

The DC bin is set to zero. After the phase noise spectrum has been derived and scaled accordingly with regards to Eq. 3.24, the inverse fourier transform is applied and the time-domain phasor given by

$$\phi(t) = \text{irfft}(\phi_f) \rightarrow LO(t) = e^{j\phi(t)}, \quad (3.25)$$

which is returned and multiplied onto the baseband signal, thus rotating each sample by a random phase offset according to the Lorentzian PSD of the PLL. Note that the same LO impairment is also applied in the receiver side, in the exact same way. To further improve the model a OFDM phase error compensation technique as presented in [9] is employed. This stems from the fact that a common phase error will be present on all subcarriers in a given transmitted/received symbol and can be removed by employing a high-pass filter given by $1 - \text{sinc}^2(fT_u)$ where T_u is the symbol length. This effectively subtracts the mean phase for every symbol. To verify the behavior of the PLL, a test setup was made with the specifications as seen in Tab. 3.2, here $S_{\phi,LF}$ sets the in band noise floor and $S_{\phi,HF}$ sets the noise floor for the VCO. For $S_{\phi,LF}$ it has been set low enough to make the phase noise plot mostly dominated by the VCO and the PLL loop, while $S_{\phi,HF}$ ensures spectrum flattening at high frequency offsets. As seen in Fig. 3.7 the response of the single sided phase noise spectrum achieves desired phase noise and loop bandwidth.

Table 3.2: PLL model parameters.

Parameter	Symbol	Value	Unit
<i>PLL</i>			
VCO phase noise	$\mathcal{L}(f_{\text{offset}})$	-90	dBc/Hz
Phase noise offset freq.	f_{offset}	1	MHz
Noise floor (LF)	$S_{\phi,HF}$	-140	dBc/Hz
Noise floor (HF)	$S_{\phi,LF}$	-160	dBc/Hz
Loop bandwidth	f_L	10	kHz

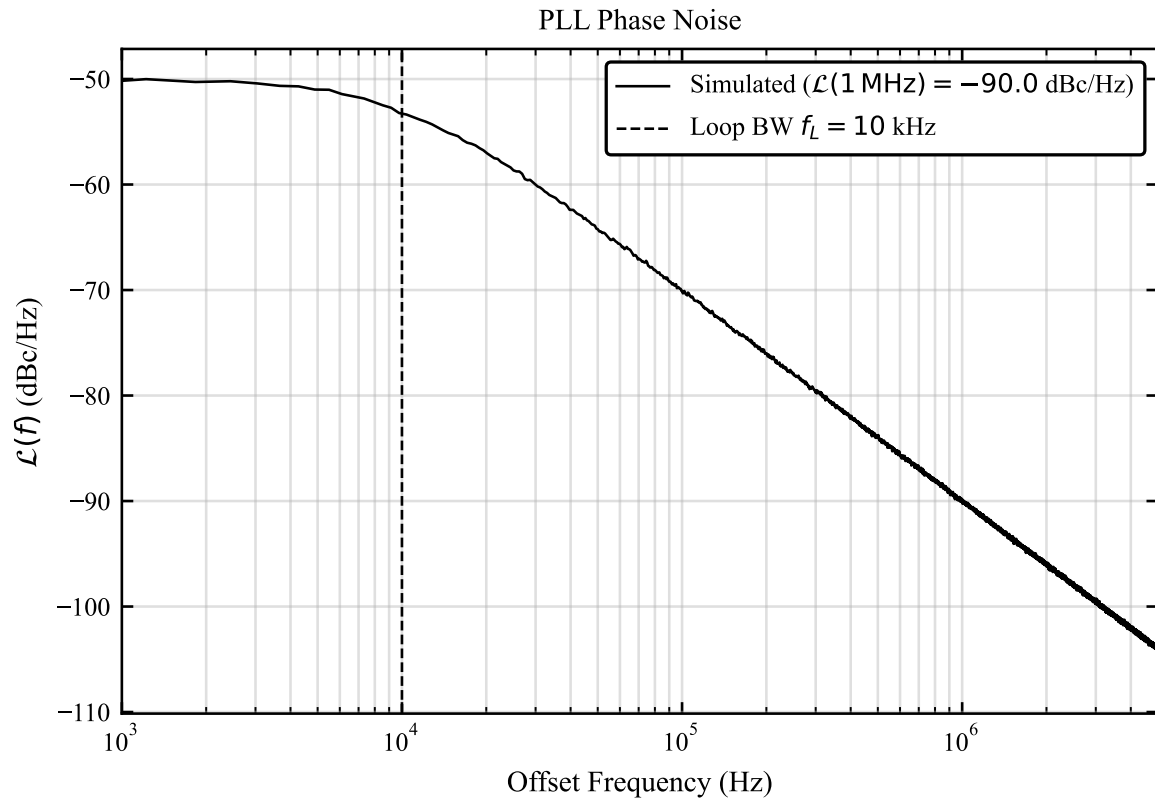


Figure 3.7: Single sided phase noise spectrum of the PLL.

3.3 Power Amplification

The power amplifier (PA) as seen in Fig. 3.8 is a fundamental component situated at the end of a transmission link before the antenna, responsible for providing power amplification i.e., driving demanding loads. The PA suffers from linearity issues given the extreme need for high dynamic range, and dramatic large signal operation. Given these high current and voltages, the PA is often seen operating close to its non-linear area. It is desired to apply the PA in its linear regime which makes linearity evaluation through compression point analysis as discussed in Sec. 2.3.3 the most important metric.

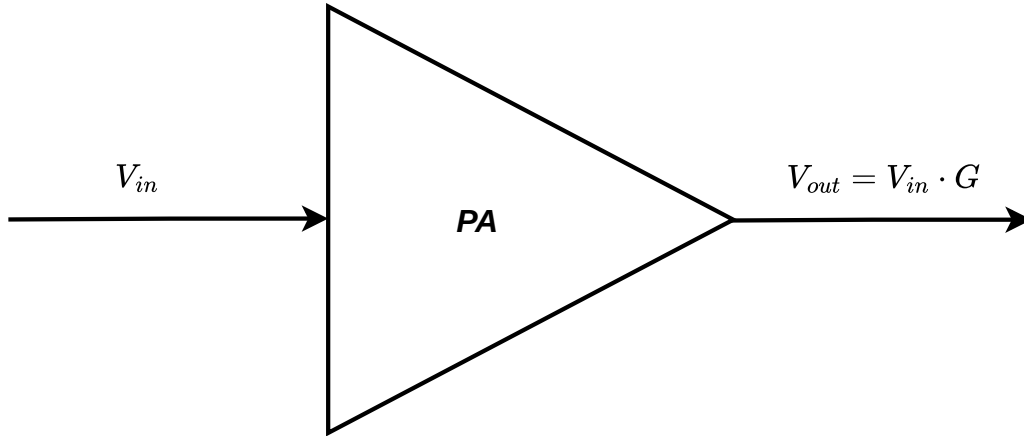


Figure 3.8: Power Amplifier block.

3.3.1 Compression Point (P1dB) modelling

As discussed in Sec. 2.3.3 and shown in Appendix. B the third order effects of the power amplifier can be modelled by the complex baseband model given by

$$y = ax - \beta|x|^2x, \quad (3.26)$$

where the beta factor is given as

$$B = (1 - c)c^2 \frac{\alpha^3}{P_{out,1dB}}, \quad (3.27)$$

where c yields the 1 dB compression point i.e., $c = 10^{-1/20}$. However this implementation proves to have some limitations. Looking at Fig 3.9a it can be seen the cubic formula provides a transfer characteristic that decreases rapidly as the PA is driven into hard saturation. This is not ideal since a PA in a real world application would have a smoother compression ratio once driven deeply into the non-linear region. Pushing the model further into the non-linear regime (not shown in Fig 3.9a) makes the amplifier start to behave non physical. To mitigate this another implementation to model the compression can be made by employing the Rapp model as given in [11].

3.3.2 PA Rapp model

The complex envelope of the input and output relation of a general amplitude and phase modulated passband signal can be written as

$$s'(t) = A(t)e^{j\phi(t)} \longrightarrow s(t) = g(A(t))e^{j(\phi(t)+\Phi(A(t)))}. \quad (3.28)$$

For solid state amplifier's the AM/PM conversion is considered small given the highly linear small signal region. This allows an analytical expression that only describes the AM/AM conversion impact given by [11]

$$g(A) = v \frac{A}{\left(1 + \left[\left(\frac{vA}{A_0}\right)^2\right]^P\right)^{\frac{1}{2P}}}, \quad (3.29)$$

Where A_0 is the limiting output amplitude, v is the small-signal gain, and p is the smoothing factor. Assuming that the instantaneous input amplitude A lies in the linear region, the linear output amplitude can be written as

$$y_{\text{lin}} = vA. \quad (3.30)$$

Let y_{out} denote the compressed output envelope amplitude. Then

$$y_{\text{out}} = \frac{y_{\text{lin}}}{\left(1 + \left(\frac{y_{\text{lin}}}{A_{\text{sat}}}\right)^{2p}\right)^{\frac{1}{2p}}}. \quad (3.31)$$

For a given 1 dB compression point, the output amplitude is 1 dB lower than the corresponding linear output amplitude, i.e.,

$$c = \frac{y_{\text{out}}}{y_{\text{lin}}} = 10^{-1/20}, \quad (3.32)$$

Thus,

$$c = \frac{1}{\left(1 + \left(\frac{y_{\text{lin}}}{A_{\text{sat}}}\right)^{2p}\right)^{\frac{1}{2p}}}, \quad (3.33)$$

and the saturation amplitude can be written as

$$A_{\text{sat}} = \frac{y_{\text{lin}}}{(c^{-2p} - 1)^{\frac{1}{2p}}}. \quad (3.34)$$

Now this can be used in a system model by specifying a given compression point and relating it to the corresponding linear output. Eq. 3.34 determines the saturation amplitude such that the PA model exhibits a 1 dB reduction in the output amplitude. The smoothness factor p determines the smoothness of the transition from the linear to the compressed region. This is visualized in Fig. 3.9c. A pseudo code of the implemented PA can be seen in Listing 3.5.

Listing 3.5: PA AM-AM nonlinearity.

```

1 def process(self, s: Signal) -> Signal:
2     x = s.x
3     # Memoryless cubic AM-AM: y = alpha*x - beta*|x|^2 * x
4     # beta_cubic is solved so that 1 dB compression occurs at P1dB_out.
5     if self.params.enable_cubic:
6         y = self.alpha * x - self.beta_cubic * (np.abs(x) ** 2) * x
7         return s.copy_with(x=y)
8
9     # Rapp Soft-Compression
10    r = np.abs(x)
11    r_lin = self.g * r
12    r_out = r_lin / (1.0 + (r_lin / self.Asat) ** (2.0 * self.p)) \
13             ** (1.0 / (2.0 * self.p))
14
15    gain_amp = np.zeros_like(r_out)
16    nz = r > 0 # guard against zero division
17    gain_amp[nz] = r_out[nz] / r[nz] # make a factor to multiply on complex
18    # samples
19
20    y = gain_amp * x
21    return s.copy_with(x=y)

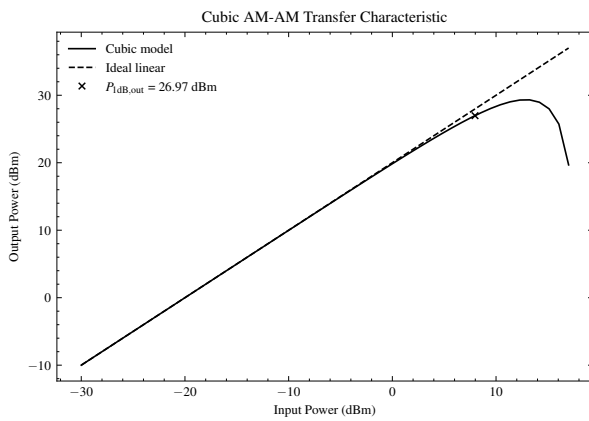
```

To evaluate the performance of the PA the model is instantiated with the parameters seen in Tab. 3.3. Looking at the plots in Fig. 3.9 the gain characteristic curves can also be seen, showing an $P_{1\text{dB},\text{in}}$ given by $P_{1\text{dB},\text{in}} = P_{1\text{dB},\text{out}} - G + 1 = 27\text{ dBm} - 20 + 1 = 8\text{ dBm}$. Another thing to consider from the discussion above, is when to use each model. As clearly seen the cubic model has problems when working in deep non-linear region, however since the Rapp model simply models a compression function, not given by rising harmonics, but rather purely mathematical kneeing, it's

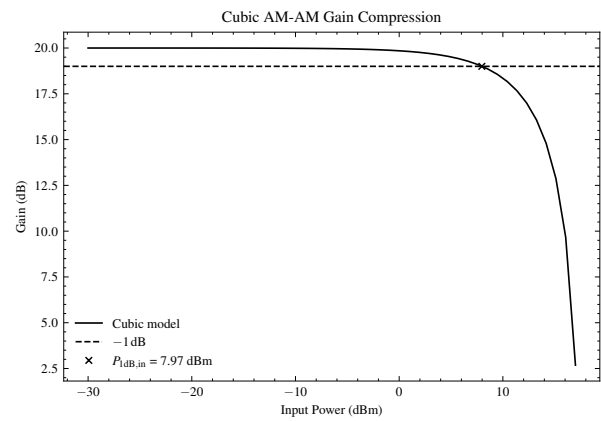
not able to capture general non-linearity, and thus harmonic distortion contributions and ultimately neglects impacts like IP3, harmonic regrowth etc. [3]. Ultimately the Rapp model was made to model Solid-State-Power-Amplifier [11] that can be pushed relatively far into compression region and still behave physical, why a model to represent this is needed however, if the overall system is driven in a comfortable linear area with modest compression the cubic model will suffice and include general non-linear contributions. Therefore the choice of each model depends on the application and test environment.

Table 3.3: Power amplifier model parameters.

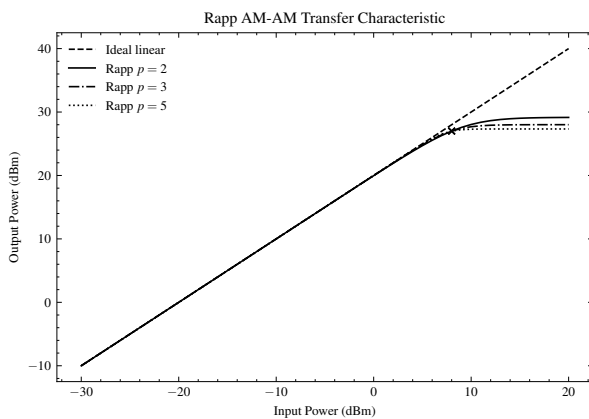
Parameter	Symbol	Value	Unit
<i>Cubic Model</i>			
Small-signal gain	G	20	dB
Output 1 dB compression	$P_{1\text{dB},\text{out}}$	27	dBm
<i>Rapp Model</i>			
Small-signal gain	G	20	dB
Output 1 dB compression	$P_{1\text{dB},\text{out}}$	27	dBm
Knee smoothness	p	2, 3, 5	–



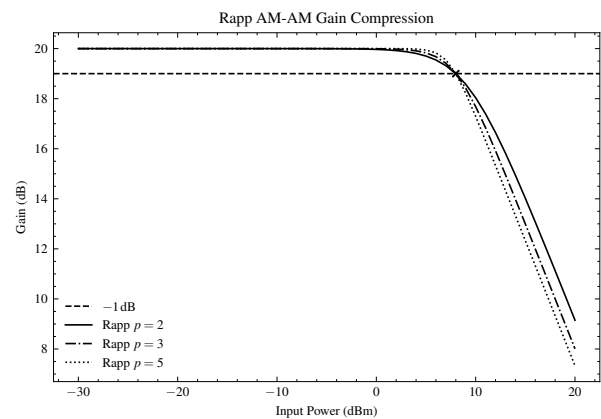
(a) Cubic model – AM/AM



(b) Cubic model – gain compression



(c) Rapp model – AM/AM



(d) Rapp model – gain compression

Figure 3.9: Comparison of the PA response for cubic and Rapp model. The left side figures show the AM/AM transfer characteristic, while the right side figures show the corresponding gain compression behavior.

3.4 Channel Modelling

Besides the individual components in the transmitter and receiver, a wireless transceiver model should also include the actual channel in which the signal is transmitted from the Tx to the Rx. As seen in Fig. 3.10 the model incorporates a path loss block and an Additive White Gaussian Noise (AWGN) channel block to model the signal propagation in free space.

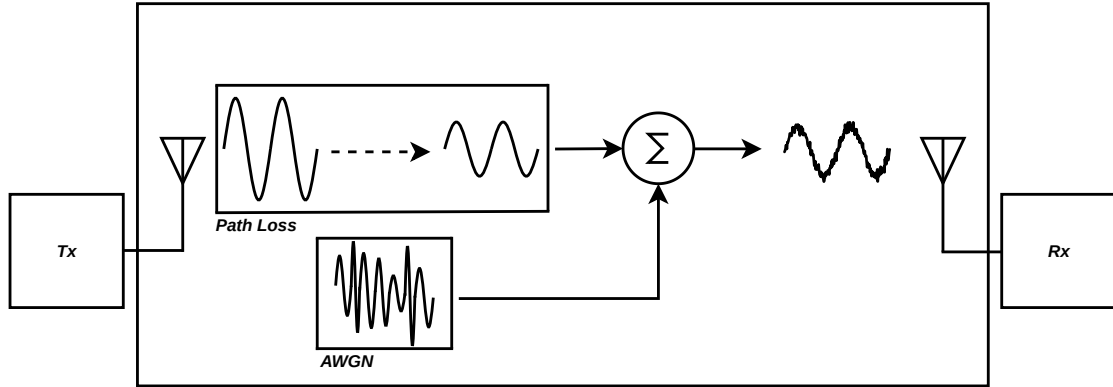


Figure 3.10: Block diagram of the modelled channel.

The class is a simple implementation of Friis transmission equation, with a warning for near-field operation if the distance is less than the Fraunhofer distance, ie., if $d < 2\lambda$. The linear antenna gain can simply be written as seen in Eq. 3.35.

$$G_t = 10^{\frac{G_{t,\text{dBi}}}{10}}, \quad G_r = 10^{\frac{G_{r,\text{dBi}}}{10}}. \quad (3.35)$$

The path loss can be modelled from Friis equation as seen in Eq. 3.36 [12].

$$G_{\text{friis}} = G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2. \quad (3.36)$$

This path loss is converted to an amplitude scaling given as $\alpha = np.sqrt(G_{\text{friis}})$ and multiplied onto the input signal. The AWGN channel is implemented by adding complex white noise to the input signal by first scaling the added noise power by the SNR requirement for the channel. This scaling of the noise power is seen in Eq. 3.37

$$P_n = \frac{P_s}{10^{\frac{\text{SNR}}{10}}}. \quad (3.37)$$

Now for a proper complex gaussian distribution the real and imaginary parts are independent and identically distributed meaning the total variance $|E[n|^2]| = 2\sigma^2$, and thus the std of the noise is given by Eq. 3.11 . The noise is generated using the `_rng.normal` function in python, which generates real-valued Gaussian noise with the given standard deviation. Finally it's multiplied onto the baseband signal. The pseudo code implementation can be seen in Listing. 3.6. To verify the behavior of the propagation channel, a testbench was setup with the settings as specified in Tab. 3.4.

Table 3.4: Channel pipeline configuration.

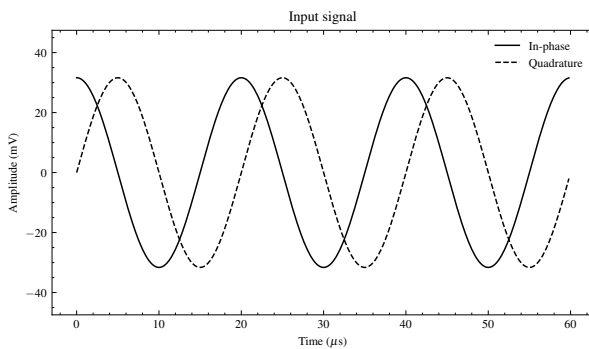
Sub-block	Parameter	Value
Path loss	Carrier frequency	1.0 GHz
Path loss	Link distance	5.0 m
Path loss	Tx antenna gain	0 dB
Path loss	Rx antenna gain	0 dB
AWGN	Target SNR	20 dB

Listing 3.6: AWGN Noise Generation.

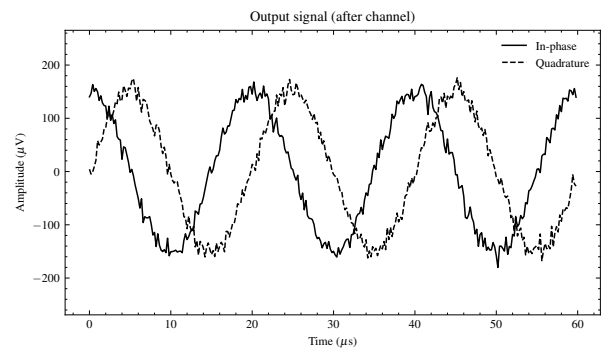
```

1  p = self.params
2  x = s.x
3
4  # Average input signal power
5  Ps = np.mean(np.abs(x) ** 2)
6
7  # Convert SNR from dB to linear
8  snr_linear = 10.0 ** (p.snr_db / 10.0)
9
10 # Required complex noise power
11 Pn = Ps / snr_linear
12
13 # Proper complex Gaussian:
14 #  $E[|n|^2] = 2\sigma^2 \Rightarrow \sigma = \sqrt{Pn/2}$ 
15 sigma = np.sqrt(Pn / 2.0)
16
17 n = (
18     self._rng.normal(0.0, sigma, size=x.shape)
19     + 1j * self._rng.normal(0.0, sigma, size=x.shape)
20 )
21
22 y = x + n

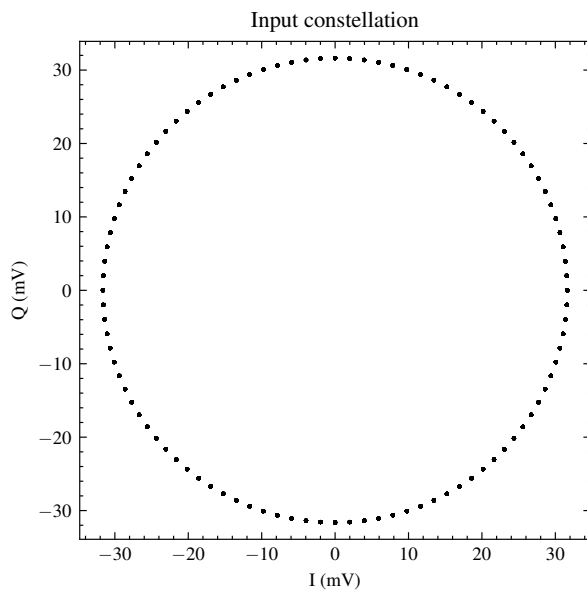
```



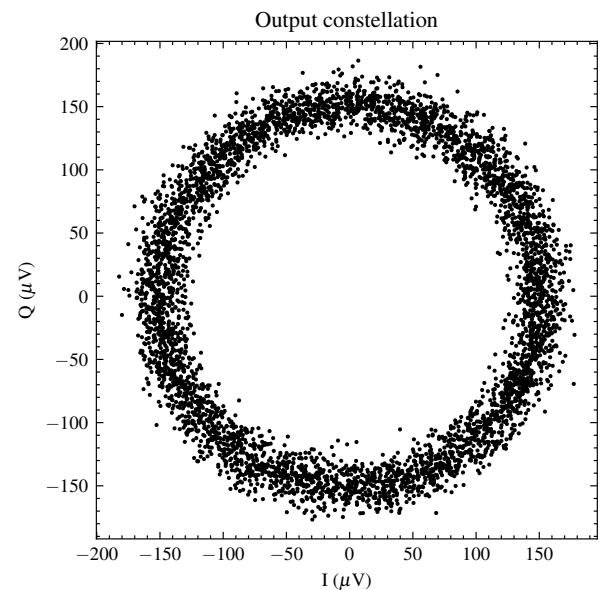
(a) Input signal to propagation channel



(b) Output signal from propagation channel



(c) Multiple periods input constellations



(d) Multiple periods output constellations

3.5 Low Noise Amplifier (LNA)

Low noise amplification is essential for amplifying weak signals in the receiver chain for further down conversion and thus digital conversion. A block diagram of the LNA can be seen in Fig. 3.12.

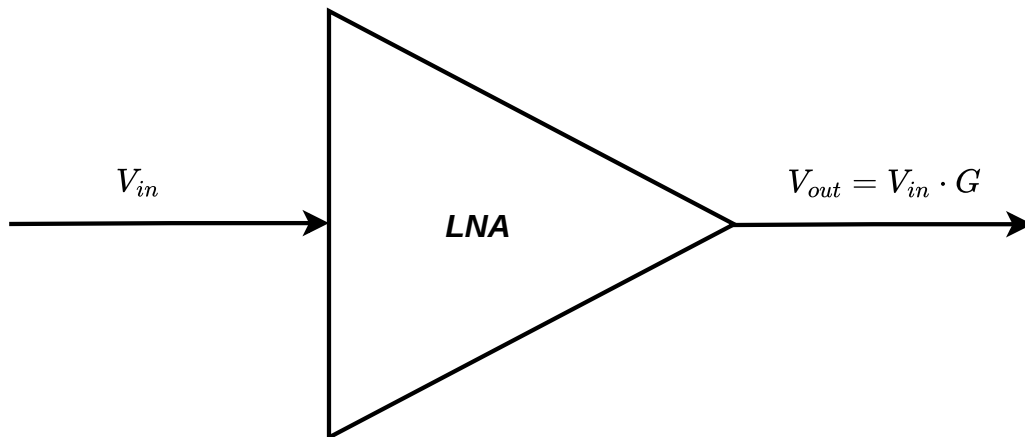


Figure 3.12: Low noise amplifier block.

3.5.1 Third Order Intercept Point

The LNA is implemented using the memoryless non-linear system model given by Eq. 2.23, where the third order intermodulation product coefficient is given as $\beta = \frac{\alpha}{2 \cdot P_{IP3}}$. Besides this the implementation is very similar to the power amplifier as given in Sec. 3.3.

3.5.2 Input Referred Noise Source

While the intercept point captures the main non-linearity's seen in the LNA, it is also critical to evaluate the noise performance, since the amplifier will be handling extremely low power signals, and therefore must suppress unwanted noise to not cover the amplified signal in additional noise. To model this an input-referred noise source as seen in Eq. 3.10 was made, in the same style as for the mixer and the AWGN channel. A pseudo code of the LNA implementation can be seen in Listing 3.5.2 ,

```

1 def process(self, s: Signal) -> Signal:
2     p = self.params
3     x = s.x
4
5     # 1. Linear voltage gain from power gain in dB
6     G = db_to_linear(p.gain_db)
7     alpha = np.sqrt(G)
8
9     # 2. Cubic AM-AM nonlinearity: y = alpha*x - beta*|x|^2 * x
10    # beta is set from input-referred IIP3 (small-signal IM3 model)
11    Pin_iip3_w = dbm_to_w(p.IP3_dbm)
12    beta = alpha / (2.0 * Pin_iip3_w)
13
14    y = alpha * x - beta * (np.abs(x) ** 2) * x
15
16    # 3. Output-referred thermal noise from noise figure
17    # N0_added = (F - 1) * k * T * G [W/Hz] (one-sided)
18    F = db_to_linear(p.nf_db)
19    k = 1.380649e-23
20    noise_psd_w_per_hz = (F - 1.0) * k * p.temp_k * G
  
```

```

21 B_hz          = s.fs_hz / 2.0
22 Pn_out_added_w = noise_psd_w_per_hz * B_hz    # E[|n|^2] per sample
23
24 # 4. Inject complex AWGN: E[|n|^2] = 2*sigma^2
25 sigma = np.sqrt(Pn_out_added_w / 2.0)
26 n      = (self._rng.normal(0.0, sigma, size=y.shape)
27           + 1j * self._rng.normal(0.0, sigma, size=y.shape))
28
29
30 y = y + n
31 return s.copy_with(x=y)

```

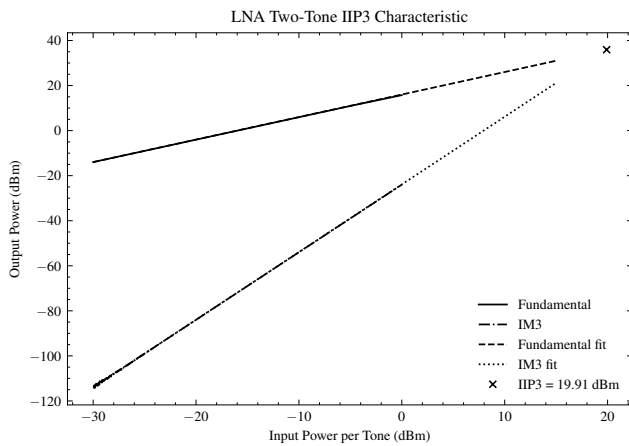
To verify the performance of the LNA, a testbench was instantiated with the parameters as seen in Tab. 3.5. To verify the third order intercept point a two tone signal $x(t) = \cos(2\pi f_1 t) + \cos(2\pi f_2 t)$ was generated with tone offsets $f_1 = 50\text{kHz}$ and $f_2 = 120\text{kHz}$, sampled at $f_s = 5\text{MHz}$. From here the per tone input power is swept from -30dBm to 0dBm by scaling the envelope directly. The output of the two tone test can be seen in Fig. 3.13a. For verifying the noise figure, a complex AWGN input is generated with an input power equal to the thermal noise floor with $T = 290\text{K}$ and a $B = 2.5\text{MHz}$. The signal is fed into the pipeline containing the LNA and the output is compared to the input to derive the noise factor from the following expression

$$F = \frac{P_{n,\text{out}}}{G \cdot P_{n,\text{in}}}, \quad \text{NF} = 10 \log_{10}(F). \quad (3.38)$$

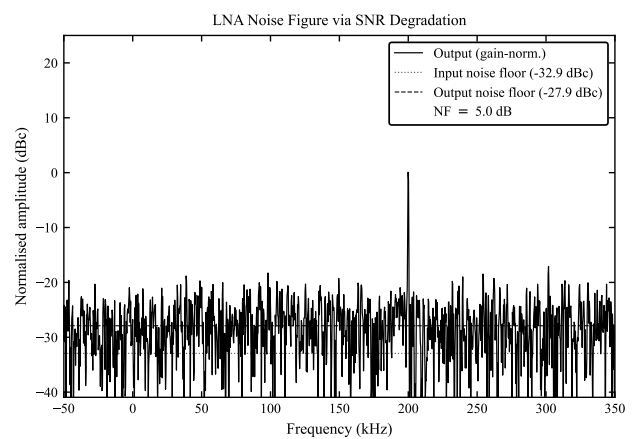
The result of the noise performance testbench can be seen in Fig. 3.13b, and as it can be seen both the intercept point and the noise performance is achieving the specified target.

Table 3.5: LNA model parameters.

Parameter	Symbol	Value	Unit
Small-signal gain	G	10	dB
Noise figure	NF	5	dB
Input-referred IP3	IIP3	20	dBm



(a) Extrapolation of the third order intercept point.



(b) Output spectrum and noise figure evaluation.

Figure 3.13: Functional verification of the LNA model.

4 | Performance Evaluation

In the following chapter the full system is evaluated to derive key specifications for the overall transceiver, which is based on the system shown in Fig. 4.1. The settings for the transceiver follow the parameters as given in [1], and summarized in Tab. 4.1. All results obtained in this work will be compared to the measurement results in the specified paper. For parameters that aren't specified in the paper however assumed, will be explained and shown clearly. As seen from Tab. 4.1, the first unit is the mixer block, where the second-order intercept point were not specified directly in the paper however, as seen in [13] similar Gilbert Cell mixer architectures, based on **0.18 μm** CMOS process can yield minimum IIP2 (worst case yield) up to +78 dBm, while commercial available mixers, for applications all the way from ISM to WLAN, such as the Analog Devices, [ADL5380](#) yields IIP2 > 65 dBm. For this simulation the default value is set to 60 dBm. For the power amplifier the cubic model is chosen as the default. Generally the reference [1] provides lots of information with regards to the performance and the individual metrics for each block in the transceiver. The individual performance of the behavioral blocks was covered in Chap. 3 and should also be counted as actual results. The input signal is generated in the same way as seen in Sec. 3.1.

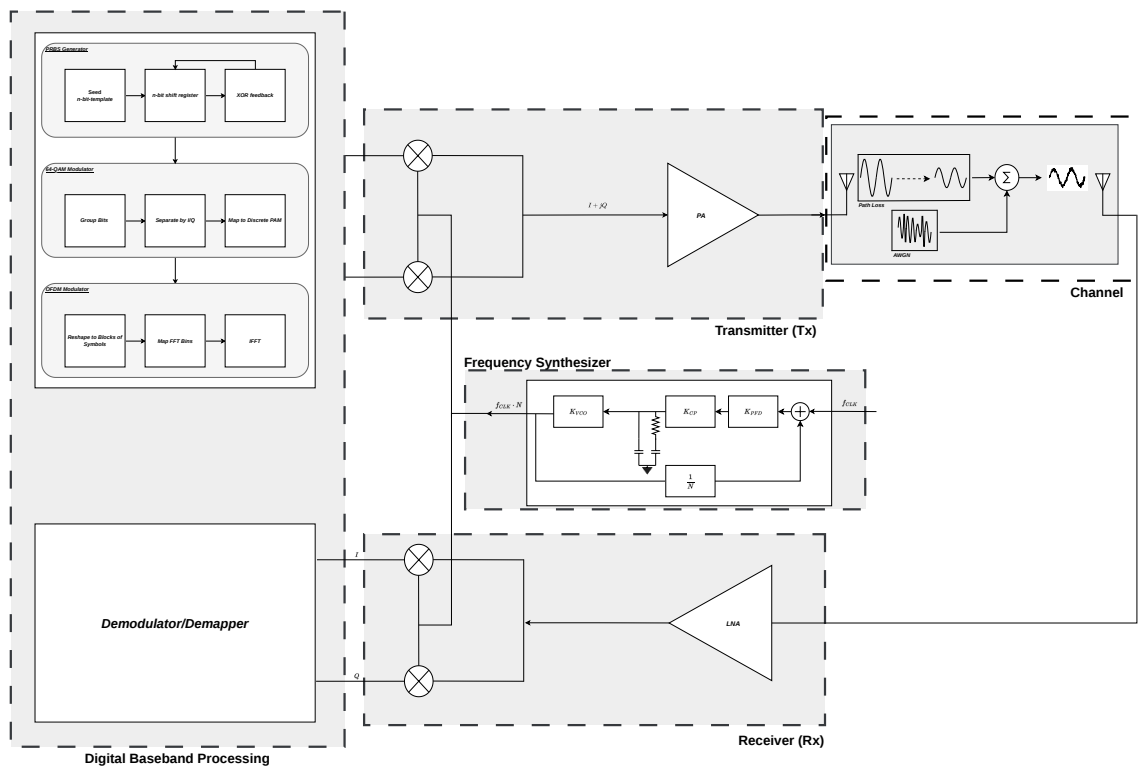


Figure 4.1: Detailed block diagram of the transceiver system.

Table 4.1: Tx–Channel–Rx pipeline configuration (Tx_channel_Rx.yaml).

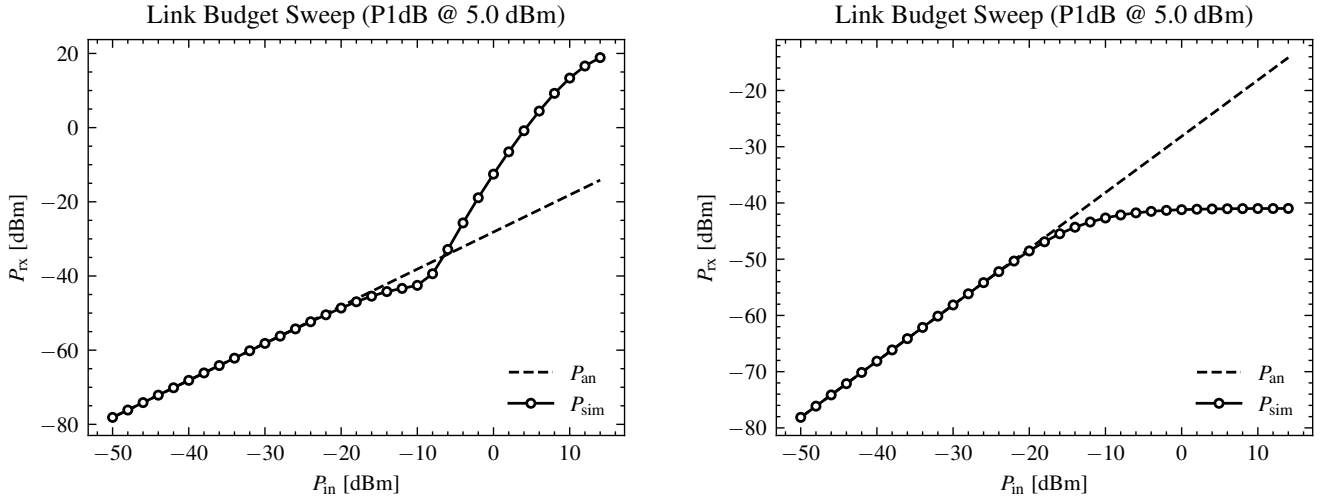
Block	Parameter	Value	Source / Note
TX Mixer + PLL	Gain	0 dB	Model reference (PGM has 27 dB range)
	IIP3	20 dBm	Paper: modulator IIP3 at max gain
	IIP2	60 dBm	Not reported; assumed
	VCO phase noise	−115 dBc/Hz @ 1 MHz	Paper: 5.15–5.35 GHz band
	LF noise floor	−140 dBc/Hz	Not reported; assumed
	Loop bandwidth	100 kHz	Paper
TX PA	Gain	20 dB	Not reported; assumed
	$P_{1\text{dB,out}}$	5 dBm	Paper
Path loss	Carrier frequency	5 GHz	5.15–5.825 GHz band
	Tx–Rx distance	2 m	Scenario choice
	TX antenna gain	2.15 dBi	Scenario assumption
	RX antenna gain	2.15 dBi	Scenario assumption
AWGN	SNR	25 dB	Scenario choice
LNA	Gain	18 dB	Paper: high-gain setting
	NF	3 dB	Paper
	IIP3	−4 dBm	Paper
RX Mixer + PLL	Gain	10 dB	Paper: quadrature demod
	IIP3	5 dBm	Paper (originally 5 dBV)
	IIP2	60 dBm	Not reported; assumed
	NF (DSB)	12 dB	Paper
	VCO phase noise	−115 dBc/Hz @ 1 MHz	Shared synthesizer with TX
	LF noise floor	−140 dBc/Hz	Not reported; assumed
	Loop bandwidth	100 kHz	Paper

4.1 Link Budget

The link budget is calculated by evaluating the received power at the Rx given by the sum of gain and losses in the system i.e.,

$$P_{RX} = P_{TX} + G_{TX} + G_{RX} - L_{path}, \quad (4.1)$$

Where P_{RX} is the received power, P_{TX} is the transmitted power, G_{TX} and G_{RX} are the transmitter and receiver antenna gains, and L_{path} is the path loss. In Fig. 4.2a, 4.2b the link budget simulation showcasing the analytical result vs the actual simulated result is showcased for both the cubic PA and Rapp PA, and as anticipated the cubic model behaves un-physical once going further into saturation region beyond $P_{1dB} = 5$ dBm. While this isn't the most advanced simulation it is however quite clear to see the impact of the path loss, as seen in Eq. 4.1.



(a) The link budget transfer function using Cubic PA model.

(b) The link budget transfer function using Rapp PA model.

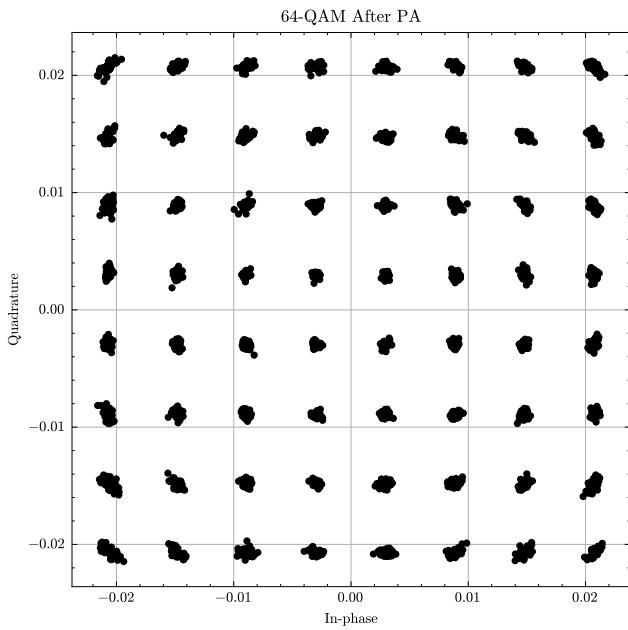
4.2 Error Vector Magnitude (EVM)

One of the most essential metrics to evaluate for a transceiver is the Error-Vector-Magnitude (EVM), since this directly relates the amount of bit errors a given system will impose. In order to compute the EVM at the transmitter and receiver, the ideal generated symbol(s) are compared to the actual transmitted symbol(s) at the PA output. From here this could be transmitted over a given channel and EVM could further be evaluated at the Rx output respectively, or a separate test setup could be made, where the ideal symbols are inferred directly to the Rx input - Both approaches will be explored and discussed. The EVM is given by Eq. 4.2

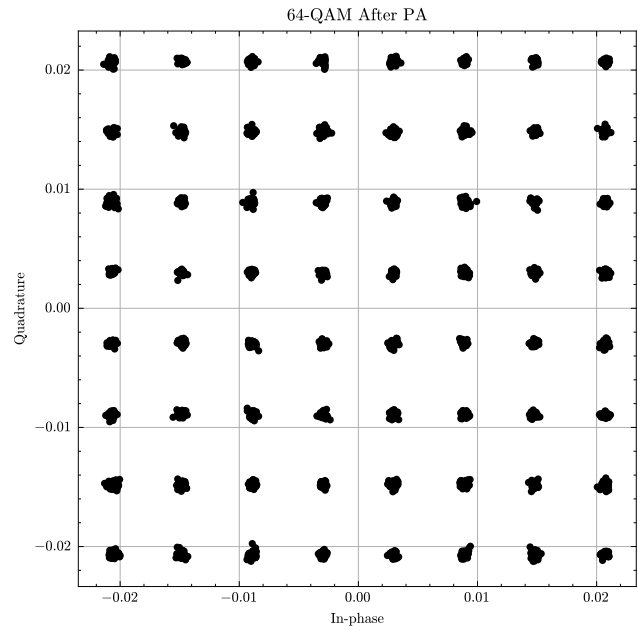
$$\text{EVM}_{\text{rms}} = \sqrt{\frac{\sum |r_k - s_k|^2}{\sum |s_k|^2}} \times 100\%. \quad (4.2)$$

4.2.1 Tx EVM

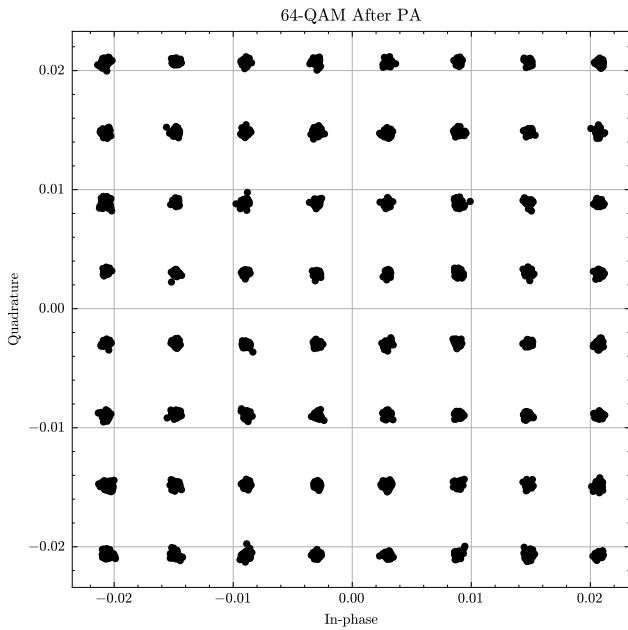
As discussed earlier in Sec. 3.2 one of the largest contributors to vectorial errors between the ideal and actual constellation point is the frequency synthesizer since phase errors directly manifest into phase rotation in the constellation. Looking at Fig. 4.3 four different constellation plots at the transmitter side can be seen. In order to understand the impact of PLL loop bandwidth and common-phase-error CPE as discussed in Sec. 3.2, the Tx was simulated with a loop-bandwidth of 100 kHz as specified in the paper and a significantly lower loop bandwidth given as 15 kHz, while the OFDM weighting function as given in [9] was toggled. As seen in Fig. 4.3a the constant phase rotation is quite evident, and seen as a constant circular rotation of all the samples, i.e., a slow varying term. Now this has a clear effect since the weighting term $1/T_u \approx 312.5$ kHz lies outside the loop bandwidth and can't be suppressed effectively. When compensating for this the constellation seen in Fig. 4.3b shows no CPE. For the actual loop bandwidth (for this simulation 100 kHz) the contribution of the common phase error becomes negligible and the output constellations can be seen in Fig. 4.3c, 4.3d. The actual RMS EVM in decibels and percent can be seen in Tab. 4.2.



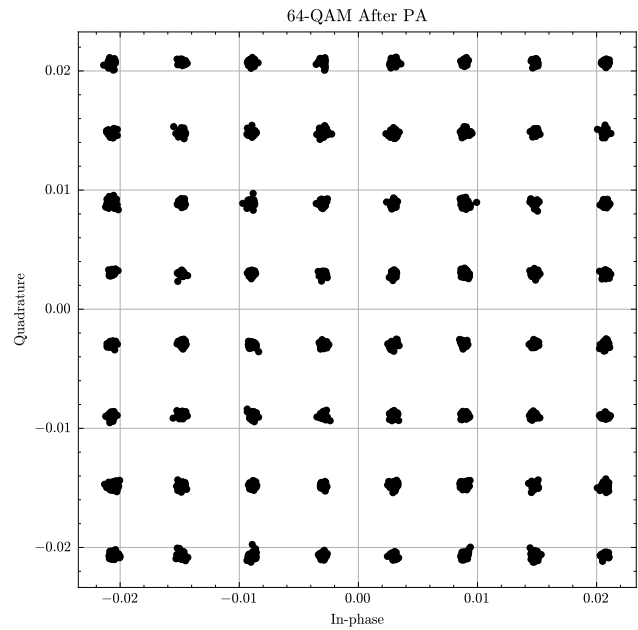
(a) Low loop bandwidth, phase error included.



(b) Low loop bandwidth, phase error removed.



(c) High loop bandwidth, phase error included.



(d) High loop bandwidth, phase error removed.

Figure 4.3: Tx 64-QAM constellations after the PA for the two PLL loop bandwidth settings, and with CPE toggling.

Table 4.2: TX RMS EVM measured at the PA output for both PLL loop bandwidths, and CPE toggling.

Configuration @ $T_{xPO} = -5$ dBm	EVM [%]	EVM [dB]
Low loop BW, with PE (a)	2.02	-33.91
Low loop BW, no PE (b)	1.40	-37.07
High loop BW, with PE (c)	1.45	-36.77
High loop BW, no PE (d)	1.38	-37.23

4.2.2 Rx EVM

In the same way as seen for the Tx, an EVM constellation can also be made for the Rx side, here this is only done for the actual used loop bandwidth as specified in Tab. 4.1. The EVM is plotted for enabled and disabled OFDM weighting function as seen in Fig. 4.4 and the actual RMS EVM is summarized in Tab. 4.3. In the following setup the EVM is evaluated for the full system, meaning it has been propagated through an AWGN channel with path loss. The SNR of the channel was given as 25 dB in accordance with Tab. 4.1. This setup was build to mimic a realistic use case, where the actual signal would be present in a noisy environment. In Tab. 4.4 the actual reported Rx EVM was evaluated by inferring the generated symbols directly to the input of the Rx with thermal noise coupled to the input to simulate a real-life setup.

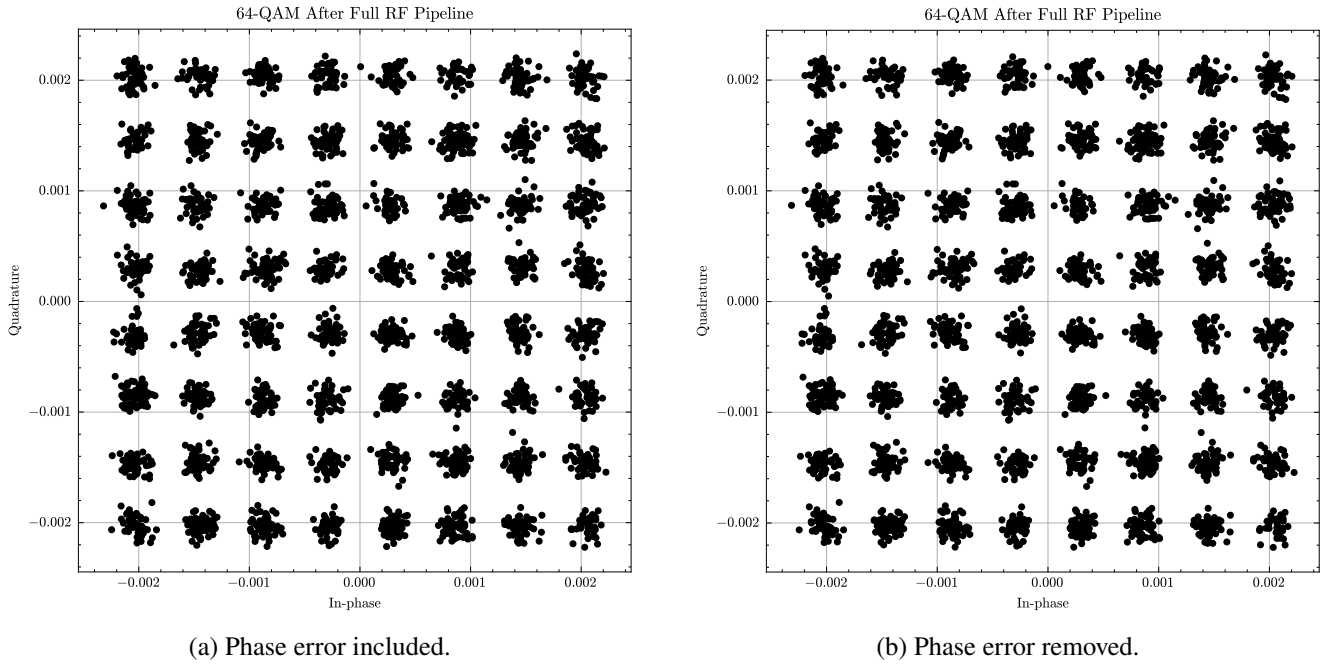


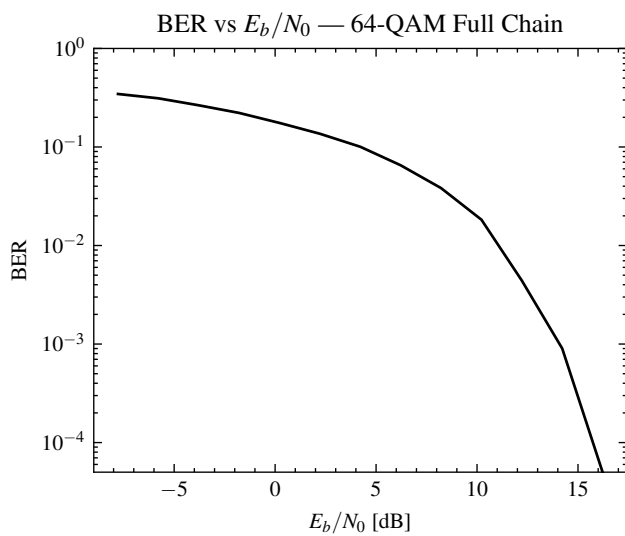
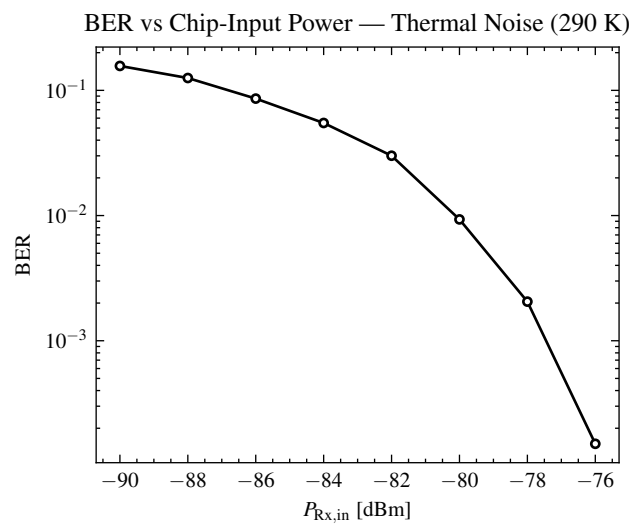
Figure 4.4: Full-chain 64-QAM constellations after RX demodulation, with and without the common phase error contribution (OFDM subcarrier weighting disabled / enabled). RMS EVM values are summarized in Table 4.3.

Table 4.3: Full-chain RMS EVM after RX demodulation, with and without the common phase error (CPE) contribution.

Configuration @ $T_{xPO} = -5$ dBm	EVM [%]	EVM [dB]
With PE (a)	5.47	-25.24
No PE (b)	5.44	-25.29

4.3 Bit Error Rate (BER) and Sensitivity

The Bit-Error-Rate (BER) is used to determine the probability of errors in the received bit stream in a wireless transmission link. To calculate the BER a series of bit streams are transmitted within a given time frame, and once received compared to the transmitted bits. From here the number of errors in the received bit stream is then divided with the total amount of transmitted bits. While the BER dramatically depends on non-linearity's such as the phase noise in the frequency synthesizer, compression point in PA and intermodulation in the mixer/LNA, these values are often fixed by design while the actual background noise in which the signal of interest is transmitted will be varying - This approach is similar to the Rx EVM simulation seen in Figs 4.4 where the EVM was evaluated after being propagated through the AWGN/Path-loss channel however, in that case the SNR was fixed to a single value. The BER proves to be very useful since it includes the impact of all elements in the communication system. The BER of the system can be seen in Fig. 4.5a as a function of E_b/N_0 . This is done by sweeping the SNR of the AWGN channel and mapping it to $E_b/N_0 = \text{SNR} - 10 \log_{10}(k)$, which yields the energy pr bit to the noise power spectral density. Furthermore the BER as a function of the receiver input power is plotted in Fig. 4.5b and from this, the Rx sensitivity for a required BER can be extrapolated.

(a) BER vs E_b/N_0 .

(b) BER vs input power (Rx).

4.4 Performance Summary

While all of the components implemented in the system model was evaluated in terms of performance metrics in Chap.3, this chapter deals with system analysis, as explained in the beginning of the chapter however, most of the tests used to evaluate the full system has already been covered, including NF simulation, compression point analysis, BER, EVM etc. To compare the results from [1] directly with the IEEE 802.11a standard and the system presented in this project, refer to Tab. 4.4.

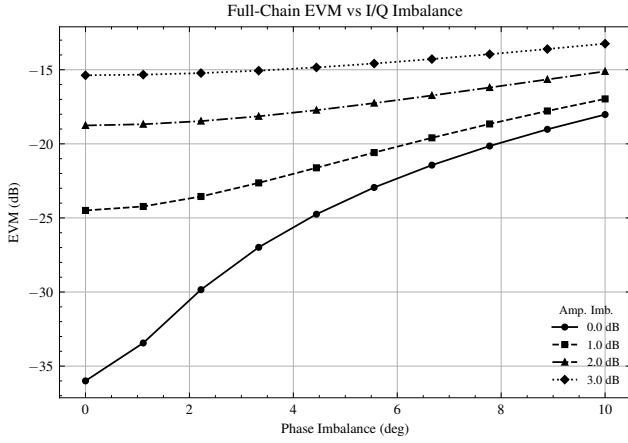
Table 4.4: Performance comparison: IEEE 802.11a standard requirements vs. the transceiver of Vassiliou *et al.* [1] vs. this work.

Parameter	802.11a Std.	Paper [1]	This Work
<i>General / Technology</i>			
Technology	–	0.18- μ m CMOS	1P6M –
Supported bands	5.15–5.35 GHz, 5.725–5.825 GHz	5.15–5.35 GHz, 5.725–5.825 GHz	Complex Base-band
<i>Receiver</i>			
RX chain NF (chip input)	–	5.2 dB	5.40 dB
RX chain gain (min)	–	20 dB	20 dB
RX path P_{-1dB} (min gain)	–	–20 dBm	–17.18 dBm
RX IIP3 (min gain)	–	–8 dBm	–7.59 dBm
Sensitivity @ 54 Mb/s ($BER = 10^{-5}$)	–65 dBm	–75 dBm	–78.15 dBm
RX EVM @ 54 Mb/s ($P_{in} = -55$ dBm)	–	–34 dB	–35.91 dB
<i>LNA</i>			
LNA gain (high/low setting)	–	18 / 10 dB	18 / 10 dB
LNA NF	–	3 dB	3 dB
LNA IIP3	–	–4 dBm	–4 dBm
<i>RX Mixer</i>			
Mixer gain	–	10 dB	10 dB
Mixer DSB NF	–	12 dB	12 dB
Mixer IIP3	–	35 dBm	35 dBm
<i>Transmitter</i>			
TX output P_{-1dB} (chip out)	–	6 dBm	5 dBm
TX EVM @ 54 Mb/s ($P_{out} = -25$ dB)	–	–33 dB	–36.77 dB
<i>Synthesizer / VCO</i>			
Architecture	–	Integer- N band PLL	wide-band 1st-order Lorentzian PLL
PLL loop bandwidth	–	500 / 100 kHz	500 / 100 kHz
Phase noise @ 1 MHz offset	–	–115 dBc/Hz	–115 dBc/Hz

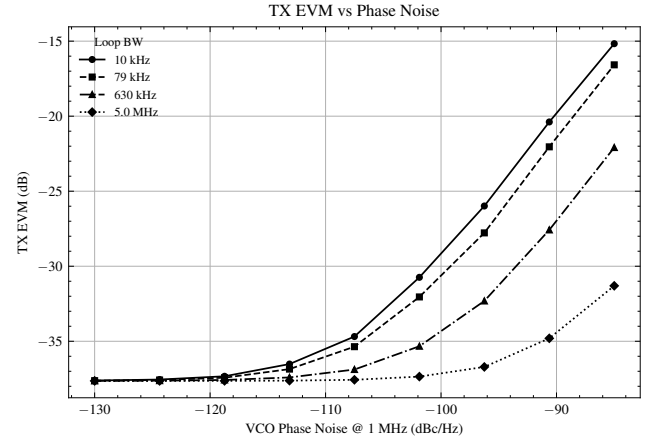
5 | Discussion and Future Work

From the results summarized in Tab. 4.4 the broad overview clearly shows that the model created in this project, provides tangible results in the same magnitude as seen in [1], one thing is the individual components behaving in the intended way, another thing is the actual system performance aligning with real hardware. Especially the EVM shows a reliable result with a deviation around 20% for the Rx EVM at 54Mb/s ($P_{in} = -55$ dBm), and a deviation around 35% for the Tx EVM at 54Mb/s ($P_o = -5$ dBm), thus making the system model predict a better performance than seen in the paper. One reason for this could be the fact the system model doesn't account for I/Q imbalance in the mixing stage with the local oscillator. This results in perfect 90° relation between the inphase and quadrature elements and equal amplitude relation, which is not realistic. To emphasize the impact of I/Q imbalance the Tx EVM at 54Mb/s ($P_o = -5$ dBm) is plotted for different constant imbalances as seen in Fig. 5.1a. As seen from the plot the penalty of amplitude imbalance yields a very dramatic change in the final EVM. While I/Q imbalance usually isn't shown directly in modern IC datasheets it is explicitly stated in [1] given as $1.5^\circ/0.2$ dB, which according the proposed system model in this report would yield a penalty in the Tx EVM of +4.87 dB yielding an EVM of ≈ -32 dB, giving the system model a worse performance than reported in the paper. While this is not directly comparable, given the fact that the on chip architecture is doing baseband processing to mitigate this imbalance, it shows a degenerate impact that clearly decreases the performance, and potentially needs to be included into the system model to get closer to actual on chip performance however, the complexity of modelling the I/Q imbalance calibration would also have to be accounted for. For further understanding the degeneration of the Rx EVM, the simulation as showcased in Fig. 4.4, was evaluated for a link distance of 2m and a channel AWGN SNR of 25 dB yielding an EVM of -25.29 dB in the best case. While this shows how sensitive a wireless link is, a better representation is seen in Fig. 4.5a where the BER is referred to the channel SNR, normalized pr bit.

Another aspect to be carefully considered is the PLL phase noise and loop bandwidth, given the fact that the EVM is extremely sensitive to the frequency synthesis and interaction with the mixer stage. As already discussed the model has 2 fundamental knobs to turn on in order to evaluate the impact of the PLL, which is the bandwidth of the loop filter and the offset phase noise contribution from the VCO. As seen in Fig.5.1b, the Tx EVM at 54Mb/s ($P_o = -5$ dBm) is evaluated as a function of the VCO phase noise at 1 MHz offset along with different loop bandwidth configurations, showing noticeable difference in the EVM for the different configurations. Its clear to see that larger bandwidth is preferred, in terms of EVM performance which intuitively makes sense, since this increases the suppression of the high frequency VCO content, impacting the phase noise dramatically however, in reality it becomes a tradeoff of tracking speed, stability, damping, die area etc. In order to understand the tradeoffs in general Type II PLL design please refer to Appendix D. It may appear counterintuitive that other components aren't evaluated to same degree as seen for the PLL, given the fact that NF, compression, intermodulation clearly has great impact on the overall system performance, however the focus is primarily on the PLL and therefore individual components are not covered in relation to system performance, further than the implicit effect it has through the system model, and the comparison table seen in Tab. 4.4.



(a) Tx EVM as a function of I/Q phase and amplitude imbalance.



(b) Tx EVM as a function of PLL phase noise and loop bandwidth.

5.1 Future Considerations

The system model under discussion comprises six main elements, namely *signal generation*, *mixing/frequency synthesis*, *power amplification*, *channel modelling*, and *low-noise amplification*. While infinite effort could be put into improving the system model, one of the main elements missing is analog/digital baseband filtering. One of the main challenges when employing transceivers in the real world, lies within the considerations of the spectral shape of the transmitted signal, and suppression of unwanted interferers in the receiver. In the 802.11a IEEE standard [5], specifications for both the spectral mask and channel rejection are given and seen summarized in Tab. 5.1. Since the OFDM symbols in frequency domain is seen as orthogonal spaced Sinc subcarriers the overall spectrum approaches a rectangular shape in the occupied bandwidth. These Sinc functions extends well beyond the frequencies of interest which is problematic since this frequency content will appear in neighboring channels (spectral leakage), thus a spectral masking is needed to protect neighboring channels. The spectral masking can be done either by analog RF filtering or digital baseband processing, where the secondary option proves to be favorable given low implementation cost and flexibility [14]. In practice a windowing function is applied to the OFDM symbol by convolution in frequency domain of a smoothens function. The smoothness function specified in IEEE 802.11a [5] is seen in Eq. 5.1 [14] where N_T is the useful OFDM symbol length and β is the roll off factor. As seen in Eq. 5.1, the implementation into the baseband signal generation would yield a smooth rise/fall transition, with a flat middle region, providing sidelobe reduction when compared to the sharper transitioning rectangular shape. While Eq. 5.1 was directly specified in the standard [5], further filter implementations such as raised cosine window as seen in Eq. 5.2 showcases a more modern approach to digital filtering enabling stronger sidelobe suppression and smoother transition. Raised cosine filtering allows compliance with more stringent standards such as 802.11p and 802.11af, thus making the system model be applicable to broader applications [14].

$$p_1(m) = \begin{cases} \sin^2 \left(\frac{\pi}{2} \left(0.5 + \frac{m}{\beta N_T} \right) \right), & -\frac{\beta N_T}{2} \leq m < \frac{\beta N_T}{2} \\ 1, & \frac{\beta N_T}{2} \leq m < N_T - \frac{\beta N_T}{2} \\ \sin^2 \left(\frac{\pi}{2} \left(0.5 - \frac{m - N_T}{\beta N_T} \right) \right), & N_T - \frac{\beta N_T}{2} \leq m < N_T + \frac{\beta N_T}{2} \end{cases} \quad (5.1)$$

$$p_2(m) = \begin{cases} \frac{1}{2} + \frac{1}{2} \cos \left(\pi \left(1 + \frac{m}{\beta N_T} \right) \right), & 0 \leq m < \beta N_T \\ 1, & \beta N_T \leq m < N_T \\ \frac{1}{2} + \frac{1}{2} \cos \left(\pi \frac{m - N_T}{\beta N_T} \right), & N_T \leq m < (1 + \beta) N_T. \end{cases} \quad (5.2)$$

As mentioned previously, analog RF filtering is also needed in the receiver in order to isolate the transmitted signal of interest. As for the digital baseband shaping, in the Tx, the channel rejection specifications for the receiver can be seen in Tab. 5.1. The required adjacent channel rejection at 54 Mb/s, meaning a neighboring channel (for instance 20MHz separation), must be at least -1 dB, while the alternate adjacent channel rejection at (40MHz separation), must be at least $+15$ dB. In other words, an OFDM Rx must be able to handle adjacent channels where the interferer

is 1 dB less than the desired signal, while for alternate channels the interferers might be up to 15 dB stronger than the desired signal – Note that the requirements get more stringent for lower data rates. To meet these requirements, some kind of filtering must happen in order to provide the needed frequency selectivity, and as seen in [1] this can be achieved with a fourth-order Chebyshev filter with 0.5 dB of ripple, allowing a adjacent channel rejection of +17 dB, and an alternate channel rejection of +22 dB 54 Mb/s. This can be implemented using the expression for a type 1 Chebyshev filter as seen in Eq. 5.3

$$|H_F(j\omega)|^2 = \frac{1}{1 + \epsilon^2 C_n^2(\omega)}. \quad (5.3)$$

Table 5.1: IEEE 802.11a [5] (54 Mbps) filtering-relevant specifications (Tx and Rx).

Block	Parameter	Value	Note
TX mask	BW	≤ 18 MHz	Spectral mask
	@ 11 MHz	≤ -20 dBr	Spectral mask
	@ 20 MHz	≤ -28 dBr	Spectral mask
	≥ 30 MHz	≤ -40 dBr	Spectral mask
RX selectivity (54 Mbps)	Adjacent channel	-1 dB	Table 91
	Alternate channel	15 dB	Table 91

Action Items for Future Implementation

The following points summarize the improvements to be implemented and discussed in the future as explained above.

- 1. I/Q imbalance modelling.** Add phase and amplitude imbalance (and a baseband calibration stage) to the mixer, as it dominates the Tx EVM.
- 2. Digital baseband windowing.** Implement the p_1 window (Eq. 5.1) and a raised-cosine window (Eq. 5.2) to meet specifications from [5].
- 3. Analog Rx channel-select filtering.** Add a 4th-order Chebyshev baseband filter (Eq. 5.3) to satisfy the adjacent and alternate channel rejection requirements from [5].

6 | Conclusion

In the following project a modular open-source Python framework for complex-baseband behavioral simulation of RF transceivers was made and demonstrated through a full 802.11a link. All the implemented components was discussed in its core implementation and the individual performance metrics was showcased to confirm operation, and finally a system level analysis was made to showcase the full transceiver performance. To compare directly a table (Tab. 4.4) was made showcasing direct comparison of the values available and relevant from [1]. The simulated performance showed close agreement with high level parameters, in particular the receive and transmit EVM at 54 Mb/s deviating by approximately by 20% and 35%, where the system predicts better performance in both cases, consistent with the missing elements such as filter implementation and I/Q imbalance. Specifically the impact of I/Q imbalance was shown in Fig. 5.1a where it was seen that amplitude imbalance causes dramatic EVM penalty. Furthermore the impact of PLL phase noise and loop bandwidth was showcased to emphasize how sensitive the system level performance is to the mentioned metrics. It is evident from Fig.5.1b that higher bandwidths provides the best performance since the loop suppresses VCO noise to a much greater extend. The original project plan included link-budget analysis and comparing different modulation schemes analytically with regards to SNR and data-rate estimations which was not included since the OFDM link simulation and implementation of the components along with proper theoretical background grew substantially in both complexity and ultimately length of the report, therefore it was prioritized to have a complete behavioral model able to take what ever input however, in this project only showcased for OFDM. All in all this system model provides a solid foundation for proper design and theoretical analysis of tradeoffs in modern wireless communication links, and can easily be expanded to include further blocks, and modulation schemes given the modular structure. In Appendix. E the link to the repository can be found along with a simple guide on how to find the documentation to get started.

A |Two-Tone Analysis

Considering following signal comprised of two tones closely separated at the same carrier frequency f_c in the passband. The signal can be written as

$$x(t) = A_1(t) \cos(2\pi f_1 t + \theta_1(t)) + A_2(t) \cos(2\pi f_2 t + \theta_2(t)). \quad (\text{A.1})$$

Now applying the golden rule for finding the complex envelope as given in [2], first $x(t)$ is written as the real part $A(t)e^{j\xi(t)}$ where $\xi(t)$ is the arguments of the sinusoid. This can be written as

$$x(t) = \text{Re} \left\{ A_1(t)e^{j(2\pi f_1 t + \theta_1(t))} + A_2(t)e^{j(2\pi f_2 t + \theta_2(t))} \right\}. \quad (\text{A.2})$$

From here the carrier term is factored out

$$x(t) = \text{Re} \left\{ \left(A_1(t)e^{j\theta_1(t)} e^{j(2\pi(f_1 - f_c)t)} + A_2(t)e^{j\theta_2(t)} e^{j(2\pi(f_2 - f_c)t)} \right) e^{j(2\pi f_c t)} \right\}. \quad (\text{A.3})$$

Now the complex envelope of the signal is given as

$$\tilde{x}(t) = A_1(t)e^{j\theta_1(t)} e^{j(2\pi(f_1 - f_c)t)} + A_2(t)e^{j\theta_2(t)} e^{j(2\pi(f_2 - f_c)t)}. \quad (\text{A.4})$$

Now applying the signal to the cubic memoryless non-linear equation given from Eq. 2.12, the following can be shown

$$y(t) = \alpha_1 \cdot \tilde{x} + \alpha_2 \tilde{x}^2 + \alpha_3 \tilde{x}^3. \quad (\text{A.5})$$

Now in order to simplify the calculation the frequencies are translated into $\omega_1 = 2\pi(f_1 - f_c)$, $\omega_2 = 2\pi(f_2 - f_c)$

Now Eq. A.4 becomes

$$\tilde{x}(t) = A_1(t)e^{j(\omega_1 t + \theta_1(t))} + A_2(t)e^{j(\omega_2 t + \theta_2(t))}, \quad (\text{A.6})$$

furthermore the phase is defined as $\phi_1(t) = \omega_1 t + \theta_1(t)$, $\phi_2(t) = \omega_2 t + \theta_2(t)$

From here this could be transformed into I/Q components in cartesian, however to equate the non-linear cubic output it's more convenient to continue in polar form. Inserting into Eq. A.5 the first order term can be written as

$$\alpha_1 \tilde{x}(t) = \alpha_1 \left(A_1(t)e^{j\phi_1(t)} + A_2(t)e^{j\phi_2(t)} \right), \quad (\text{A.7})$$

the second order term can be written as

$$\begin{aligned} \alpha_2 \tilde{x}^2(t) &= \alpha_2 \left(A_1(t)e^{j\phi_1(t)} + A_2(t)e^{j\phi_2(t)} \right)^2 \\ &= \alpha_2 \left(A_1^2(t)e^{j2\phi_1(t)} \right. \\ &\quad \left. + A_2^2(t)e^{j2\phi_2(t)} \right. \\ &\quad \left. + 2A_1(t)A_2(t)e^{j(\phi_1(t) + \phi_2(t))} \right), \end{aligned} \quad (\text{A.8})$$

the third order term can be written as

$$\begin{aligned} \alpha_3 \tilde{x}^3(t) &= \alpha_3 \left(A_1(t)e^{j\phi_1(t)} + A_2(t)e^{j\phi_2(t)} \right)^3 \\ &= \alpha_3 \left(A_1^3(t)e^{j3\phi_1(t)} + 3A_1^2(t)A_2(t)e^{j(2\phi_1(t) + \phi_2(t))} \right. \\ &\quad \left. + 3A_1(t)A_2^2(t)e^{j(\phi_1(t) + 2\phi_2(t))} + A_2^3(t)e^{j3\phi_2(t)} \right). \end{aligned} \quad (\text{A.9})$$

Now following the standard non-linear cubic formula for defining the non-linearity, it is seen that the usual difference IM3 products i.e., $2\omega_1 - \omega_2$, $2\omega_2 - \omega_1$ does not appear. This issue comes from the definition of Eq. A.4. This comes from the fact that a real passband signal that is converted to its analytical signal $s^+(t) = s(t) + j\hat{s}(t)$ contains no negative frequency components, and for the passband signal centered at f_c the analytical signal becomes

$$s^+(t) = \tilde{s}(t)e^{j2\pi f_c t}. \quad (\text{A.10})$$

Now effectively a real signal includes $+\omega_c$ and $-\omega_c$ terms, however using the complex enveloping and the standard cubic non-linear formula doesn't capture the mixing of these components. In [15] a quasi-memoryless PA has been modelled using the polynomial model as given in the following expression to capture the non-linear effects

$$z(t) = x(t) \sum_{k=0}^K a_{2k+1} |x(t)|^{2k}. \quad (\text{A.11})$$

Where $x(t)$ is the complex baseband signal. Interestingly it's seen that the complex conjugate is included in this expression through the fact that $x(t)|x(t)|^2 = x(t)x(t)x^*(t) = x^2(t)x^*(t)$. This allows the cubic baseband to be written as

$$y(t) = a_1 x(t) + a_3 x^2(t)x^*(t). \quad (\text{A.12})$$

Now the 2-tone test can be carried out again:

$$x(t) = A_1 e^{j\omega_1 t} + A_2 e^{j\omega_2 t}, \quad (\text{A.13})$$

$$x^*(t) = A_1 e^{-j\omega_1 t} + A_2 e^{-j\omega_2 t}, \quad (\text{A.14})$$

$$\begin{aligned} x(t)|x(t)|^2 &= (A_1 e^{j\omega_1 t} + A_2 e^{j\omega_2 t}) \\ &\times \left(A_1^2 + A_2^2 + A_1 A_2 e^{j(\omega_1 - \omega_2)t} + A_1 A_2 e^{j(\omega_2 - \omega_1)t} \right), \end{aligned} \quad (\text{A.15})$$

$$\begin{aligned} x(t)|x(t)|^2 &= \left(A_1^3 + 2A_1 A_2^2 \right) e^{j\omega_1 t} + \left(2A_1^2 A_2 + A_2^3 \right) e^{j\omega_2 t} \\ &+ A_1^2 A_2 e^{j(2\omega_1 - \omega_2)t} + A_1 A_2^2 e^{j(2\omega_2 - \omega_1)t}. \end{aligned} \quad (\text{A.16})$$

Assuming a standard two-tone test with equal amplitudes ($A_1 = A_2 = A$), the fundamental and intermodulation (IM) products can be summarized in Tab.A.1.

Term	Calculation	Result	Frequency
Fundamental 1	$2A^2 \cdot A e^{j\omega_1 t} + A^2 e^{j(\omega_1 - \omega_2)t} \cdot A e^{j\omega_2 t}$	$3A^3 e^{j\omega_1 t}$	ω_1
Fundamental 2	$2A^2 \cdot A e^{j\omega_2 t} + A^2 e^{j(\omega_2 - \omega_1)t} \cdot A e^{j\omega_1 t}$	$3A^3 e^{j\omega_2 t}$	ω_2
IM3 Product 1	$A^2 e^{j(\omega_1 - \omega_2)t} \cdot A e^{j\omega_1 t}$	$A^3 e^{j(2\omega_1 - \omega_2)t}$	$2\omega_1 - \omega_2$
IM3 Product 2	$A^2 e^{j(\omega_2 - \omega_1)t} \cdot A e^{j\omega_2 t}$	$A^3 e^{j(2\omega_2 - \omega_1)t}$	$2\omega_2 - \omega_1$

Table A.1: Intermodulation products of the complex envelope $|x|^2 x$ for $A_1 = A_2 = A$.

Now the time domain response of the low-noise amplifier can be described using the polynomial baseband model. Accounting only for the dominant third-order nonlinearity, the amplifier output can be written as

$$y(t) = \alpha x(t) - \beta x(t)|x(t)|^2. \quad (\text{A.17})$$

Now the linear small signal gain is given by the first order term

$$y(t) \approx \alpha x(t). \quad (\text{A.18})$$

The third-order intercept point is defined as the input level at which the extrapolated power of the fundamental and the IM3 product become equal. Using the amplitudes of the corresponding terms, the third order coefficient can be found from the following

$$|\alpha A| = |\beta A^3|. \quad (\text{A.19})$$

Solving for β results in

$$\beta = \frac{\alpha}{A^2}. \quad (\text{A.20})$$

P_{IIP3} represents the average power at the intercept point. For a signal with peak amplitude A , the power is defined as $P = A^2/2$, which implies $A^2 = 2P_{\text{IIP3}}$. Substituting this into the expression for β , the third-order coefficient can be expressed as

$$\boxed{\beta = \frac{\alpha}{2P_{\text{IIP3}}}}. \quad (\text{A.21})$$

B |1 dB Compression Point Derivation

The complex baseband memoryless cubic expression is given as

$$y = \alpha x - \beta |x|^2 x, \quad (\text{B.1})$$

$$(\text{B.2})$$

and the absolute value can be written as

$$|y| = |x|(\alpha - \beta |x|^2). \quad (\text{B.3})$$

At 1 dB compression:

$$|y| = c \cdot |y_{\text{lin}}| = c\alpha |x|, \quad c = 10^{-1/20} \quad (\text{B.4})$$

Equating:

$$|x|(\alpha - \beta |x|^2) = c\alpha |x|. \quad (\text{B.5})$$

Cancel $|x|$:

$$\alpha - \beta |x|^2 = c\alpha. \quad (\text{B.6})$$

Solve for β :

$$\beta = \frac{(1 - c)\alpha}{|x|^2}. \quad (\text{B.7})$$

Now for complex baseband signal the power is given by $P = E|y|^2$, meaning the P_{out} can directly be written as:

$$P_{\text{out,ideal}} = \alpha^2 |x|^2. \quad (\text{B.8})$$

And with a 1 dB decrease in the output characteristics yields:

$$P_{\text{out,1dB}} = c^2 \alpha^2 |x|^2. \quad (\text{B.9})$$

Thus:

$$|x|^2 = \frac{P_{\text{out,1dB}}}{c^2 \alpha^2}. \quad (\text{B.10})$$

This can then be inserted into (B.7)

$$\beta = (1 - c)\alpha \cdot \frac{c^2 \alpha^2}{P_{\text{out,1dB}}}, \quad (\text{B.11})$$

This yields the beta factor as:

$$\boxed{\beta = (1 - c) c^2 \frac{\alpha^3}{P_{\text{out,1dB}}}.} \quad (\text{B.12})$$

C |IEEE Std 802.11 (2012)

In this chapter a analysis of the IEEE 802.11 standard with regards to *Wireless LAN Medium Access control and Physical Layer Specification's* is made.

C.1 OFDM PHY Specification

In the IEEE 802.11 the Physical Layer Convergence Procedure (PLCP) provides the interface between the MAC layer and the physical layer (PHY). The full transmitted PHY packet is called the PPDU which stand's for *PLCP Protocol Data Unit* and stems from the MAC-delivered payload. The PPDU contains a preamble signal header and the actual data payload as seen in Fig. C.1. In [5], an overview of the PPDU encoding process is specified in chapter 18.3.2.2. In the full standard, the DATA field is formed from the *SERVICE*, *PSDU*, *tail*, and *pad* fields, followed by scrambling, channel coding, interleaving, and modulation. In the present system model, the full PLCP framing is not implemented however, emphasis is placed on generating a pseudo random binary payload for the modulation and OFDM chain in order to ensure correct testing environment. This allow's to simulate realistic unpredictable conditions.

$$b_{out}[n] = b_{in}[n] \oplus s[n] \quad (C.1)$$

Where $s[n]$ is a pseudo-random bit. The pseudorandom bit's are provided by a generator polynomial (bit shift register) $S(x)$, and for pseudo-random binary sequence of order 7 the generator polynomial in *802.11* is given as

$$S(x) = x^7 + x^4 + 1 \quad (C.2)$$

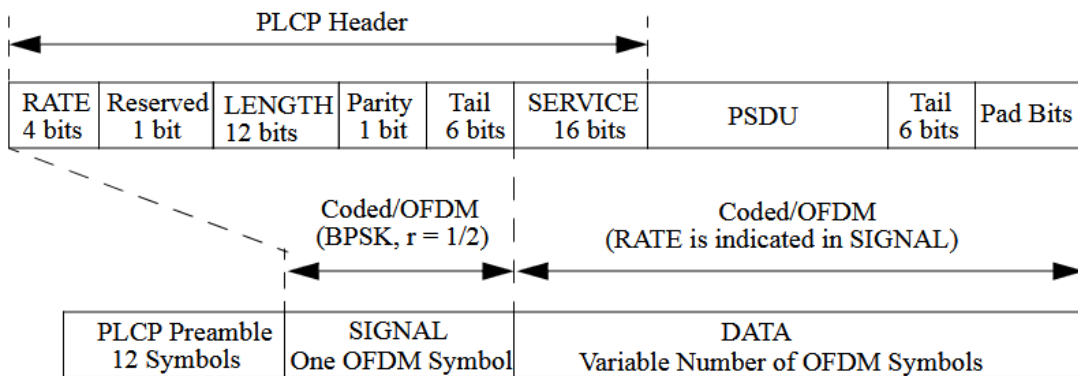


Figure C.1: PPDU Frame Format, taken from *Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications*.

C.2 Code Implementation PRBS

A pseudo random binary sequence (PRBS) generator was implemented as a functional block in accordance to what is seen in Fig. C.2. The block generates binary payloads and is implemented for both PRBS7 and PRBS15. The block takes an input PRBS order, a non zero initial seed integer that maps to binary value, and the overall number of output bits to return.

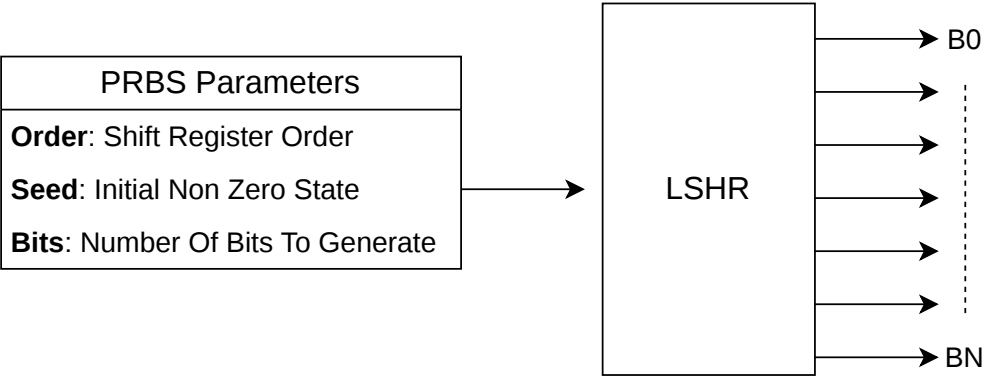


Figure C.2: Block Diagram of the Code Implementation of the PRBS.

D |Type II PLL Design Considerations

In the following example, a walkthrough of general design considerations for a Type II PLL is shown. This will be following the PLL architecture as seen in Fig. 3.6. In specific an overview of the system analysis will be presented along with discussion about VCO and loop filter design considerations. Divider circuit, and phase difference detection will not be discussed.

D.1 System Analysis

To set the requirements for individual components in the PLL, it's important to understand what is needed. Among this, is the frequency range, carrier spacing, phase noise performance, power consumption etc. One of the most fundamental parameters to determine is the division ratio, based on the channel spacing available from the reference. It's seen from Fig. 3.6 that the divider chain provides an integer division ratio, meaning the factor N would have a minimum resolution of $N \pm 1$, i.e., the division ratio can change by exactly the reference frequency, or a multiple of the reference frequency. In order to understand the tradeoff of this, let's assume that an application specifies a frequency range of 5–6 GHz (close to what is seen in [1]), now with a 20 MHz channel spacing containing useful information would require a division ratio given by Eq. D.1 and for substituting $f_0 = 5$ GHz, $\Delta f = 20$ MHz and $K = 50$ yields a sequence of division ratios given in Eq. D.2

$$N_k = \frac{f_{out,k}}{f_{ref}} = \frac{f_0 + k\Delta f}{f_{ref}}, \quad (D.1)$$

$$N_k \in \{250, 251, 252, \dots, 299, 300\}. \quad (D.2)$$

Now the division ratio directly impacts the phase noise contribution and this can be approximated by an phase noise multiplication factor as shown in Eq. D.3 as given in [16].

$$20 \log_{10}(N_k) \in [47.96, 49.54] \text{ dB}. \quad (D.3)$$

A division ratio of several hundreds is achievable for an integer PLL, and might be appropriate for frequency hopping in WLAN area in accordance with the IEEE 802.11a standard, however this is at the edge of what is possible. Pushing the system requirements to the extreme and assuming a carrier frequency at 2.4 GHz ISM band, and a carrier spacing of 1 kHz for ultra fine tuning, then the required division ratio becomes

$$N = \frac{2.4 \text{ GHz}}{1 \text{ kHz}} = 2,400,000$$

yielding a phase noise penalty proportional to $20 \log(N) \approx 127.6$ dB. Now given the fact that charge-pump PLLs behaves like a sampled-data feedback system with sampling frequency at $f_s = f_{ref}$ the PLL can become unstable if the loop bandwidth becomes too large in bandwidth when compared to the sampling frequency. A rule of thumb as specified in [17], determines the loop bandwidth to be reduced $f_{ref}/10$. Now the closed-loop characteristics of the type-II PLL is given as

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \quad (D.4)$$

and for a damping factor of $\zeta \approx 1$ the open loop unity gain BW can be written as $\omega_u \approx 2.1\omega_n$ giving

$$\omega_u = 2.1\omega_n = \frac{\omega_{ref}}{10}. \quad (D.5)$$

Now for a frequency spacing of 20 MHz implies a loop bandwidth of around 2 MHz where as the a frequency spacing of 1 kHz would imply a loop bandwidth of around 100 Hz. A loop bandwidth of 100 Hz is not feasible, since the PLL would be extremely slow, and have poor phase noise suppression, however the extreme does showcase the limitations and many variables to navigate in, when just looking at the selected frequency and spacing resolution. If an application required a fine channel spacing at high frequency, fractional PLLs are generally used since the division ratio can take the form of a non-integer value, as seen in Eq. D.6.

$$f_{out} = \left(N + \frac{F}{M}\right) \cdot f_{ref}. \quad (D.6)$$

D.2 VCO Design

One of the most challenging aspects in PLL design is the VCO, since this is the component responsible for actually delivering the synthesized frequency. The VCO can be implemented in various ways, depending on the requirements for the application, where some of the most widely used VCO's are ring oscillators and LC tank based oscillators, and the main benefits of each is summarized in Tab. D.1. As seen throughout this project the phase noise of the PLL is a critical parameter for meeting overall system performance, and given the loop dynamics of a PLL the VCO contributes largely to the overall phase noise. For a general differential LC oscillator as shown in Fig. D.1 the phase noise contribution can be described by Lessons's equation [18] as seen in Eq. D.7

$$L(\omega_m) = \frac{4FkTR_p}{V_{rms}^2} \left(\frac{\omega_o}{2Q\omega_m} \right)^2, \quad (D.7)$$

where F is the oscillator noise factor, V_{rms} is the rms oscillation amplitude, R_p is the equivalent parallel resistance of the resonator, ω_m is the offset frequency and Q is the loaded quality factor of the resonator. Generally the phase noise contribution can be simplified to an inverse proportionality of the bias point and LC tank quality factor.

$$L(\omega_m) \propto \frac{1}{I_{bias}^2 R_p Q^2}. \quad (D.8)$$

It is tempting to conclude, when looking at Eq. D.8, that the bias point of the oscillator simply can be increased in order to decrease the phase noise, however the voltage swing increases accordingly, and can cause breakdown of the active device while the power consumption of the VCO increases. Even though there is a clear tradeoff it can generally be said that high voltage swing indeed is needed in the VCO tank to decrease the overall phase noise. Since the voltage swing will be depending on the parallel equivalent resistance, a center tapped inductor can be used since the equivalent resistance seen looking from the Drain/Collector up into the LC inductor will be decreased with $\frac{1}{\text{tapping ratio}}$ [18].

Ring Oscillator	LC Tank Oscillator
• Small area • High locking range • Easy to integrate in CMOS • Wide tuning range	• Better phase noise performance • Low power at high frequency operation • High operational frequency capability • Higher spectral purity due to resonant tank

Table D.1: Comparison of Ring Oscillator and LC Tank Oscillator [16]

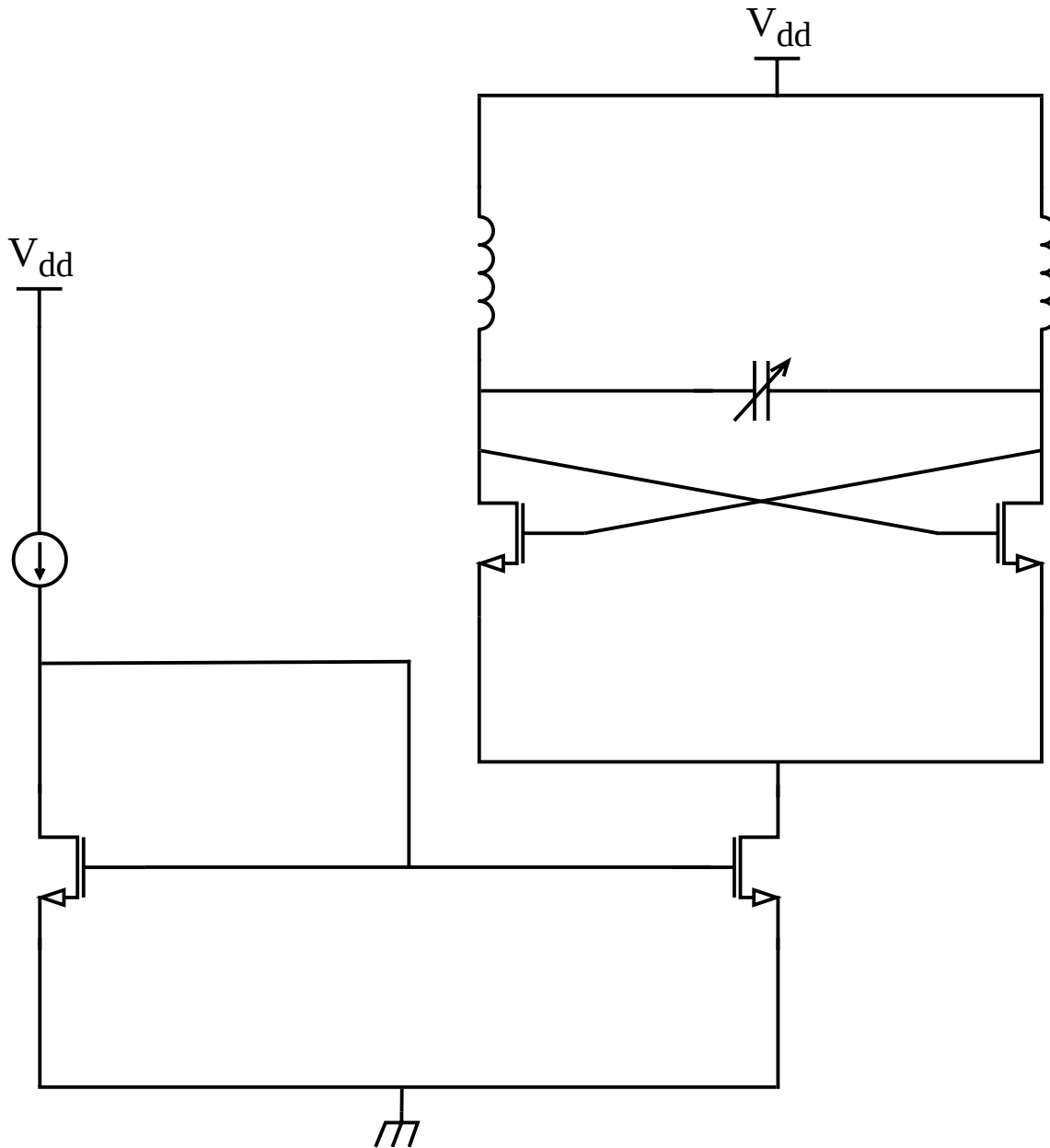


Figure D.1: Differential LC oscillator.

D.3 Loop Filter

As stated previously in the report the loop filter is responsible for providing the required control voltage for the VCO, through a low pass filter by extracting the average phase error from the PFD. For a charge-pump PLL as seen in Fig. 3.6 the loop filter pole for a strongly overdamped PLL can be seen in Eq. D.9

$$f_l \approx \frac{I_{CP} K_{VCO} R_1}{2\pi N}, \quad (\text{D.9})$$

Where I_{CP} is the charge-pump current and $K_{VCO} = \frac{df_{VCO}}{dV_{ctrl}}$ is the VCO gain, and R_1 is the filter resistance and N is the division ratio. From the Lorentzian phase noise model as shown in Eq. 3.16 and implemented in the system model, it was seen that the phase noise contributions can be segmented into, in-band noise (PFD, charge-pump etc.) and VCO noise, where the loop bandwidth trade-offs the suppression of each region, i.e., larger bandwidth more

in-band noise, less VCO noise and vice versa. Now given D.9, it can be seen that I_{CP} is set by the charge-pump design, K_{VCO} is set by VCO and system constraints, and N is set by the input/output relation, this fundamentally leaves very little space for tuning these values to hit a specific loop bandwidth, therefore the filter resistance becomes the main tuning knob. It was shown in [10] that evaluating the Lorentzian spectrum yields an "optimum" sizing of the filter resistance given by Eq. D.10

$$R_{1,opt} \approx \frac{2\pi N}{I_{CP}K_{VCO}} \sqrt{\frac{\alpha}{S_{LF}}}. \quad (D.10)$$

It has to be noted that this is an optimization focusing on RMS jitter optimization, and ignores other parameters such as settling time, spur suppression etc. To address these additional performance metrics the loop filter capacitors may be considered where the capacitor in series with $R_{1,opt}$ sets the damping and the additional shunt capacitor act suppresses reference spurs.

E |Github Repository

The repository for the github project can be found by simply clicking [here](#). The repository contains all the source packages containing all the model scripts along with additional helpers and system pipeline. Furthermore the full documentation can be found when clicking [here](#). The documentation contains all examples used in the project, along with a detailed explanation on how the pipeline interprets the blocks, and how to modify the system chain by inserting and removing blocks and modifying parameters in realtime.

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